



Deep multivariable spatial attention CNN-based spatial grid NWP error correction for accurate one-day-ahead photovoltaic power forecast

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Abstract

Accurate day-ahead solar photovoltaic (PV) power forecasting is crucial for optimizing solar energy utilization. Traditional methods typically involve interpolating numerical weather prediction (NWP) data to a specific location, often overlooking spatial information from neighbouring areas. Entropy serves as an effective tool for quantifying forecasting uncertainty. In this study, we introduce transfer entropy to capture the information gain from incorporating NWP data from spatially adjacent regions, and propose the concept of spatial grid NWP correction. A deep spatial multivariable attention convolutional neural network-based framework for spatial grid NWP error correction is then proposed for day-ahead PV forecasting. Experiments conducted on three years (2017–2019) of actual data from two PV stations in Belgium demonstrate the method's effectiveness. For the two stations, the proposed approach yields a normalized mean absolute error (nMAE) of 2.08% and 2.22%, and a normalized root mean squared error (nRMSE) of 4.45% and 4.81%, respectively. Compared to physical model chain without NWP correction, the proposed method reduces nMAE by 23.97%–24.79% and nRMSE by 15.54%–20.19%. Furthermore, comparison with eight conventional NWP correction methods further verifies its superior performance. Further experiment at a PV station in San Diego, USA, verifies that the proposed method can outperform conventional methods in different geographical regions and weather patterns.

Keywords Photovoltaic power forecast · Solar energy · Deep spatial attention convolutional network · Probabilistic forecast · Numerical weather prediction

Introduction

Nowadays, the overuse of fossil fuels has caused serious air pollution and climate change (Fan et al. 2023; Gu et al. 2023). This makes the development of renewable energy become increasingly important (Kim et al. 2024). Developing solar photovoltaic (PV) power is one of the most important ways to tackle climate change and achieve carbon

neutrality (Wang et al. 2023a; Liu et al. 2023). Due to the variability in solar irradiance resulting from cloud motions, PV power exhibits greater uncertainties than conventional coal-fired power (Collins et al. 2018; Prol et al. 2024). Forecasting is one of the most important methods to address the uncertainties of PV power, playing an important role in ensuring the stability and economic operation of power systems (Wang et al. 2023b; Said and Alanazi 2023).

According to the demands of electricity grid dispatching, the forecast horizon of PV power can be mainly divided into intra-day forecast (Xia et al. 2024) and day-ahead forecast (Wang et al. 2024). Intra-day forecast refers to the forecast within the day, while day-ahead forecast refers to the forecast for the future one to several days. One-day-ahead forecasts can offer valuable insights for day-ahead electricity market trading, spinning reserves scheduling and demand response activity arrangement for the following day, and thus is essential for enhancing the operational economics of electricity grid Weixing et al. 2023; David et al. 2021; Ahmed and Khalid 2019; Bai et al. 2023.

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Numerical Weather Prediction (NWP) is the commonly used method for day-ahead PV power forecast. NWP predicts the future weather condition through physical mechanisms and usually has better forecasting performance for long forecasting horizons than common-used purely data-driven methods. Currently, there are many mature NWP products, such as the High Resolution (HRES) model from European Centre for Medium-Range Weather Forecasts (ECMWF) (Balsamo et al. 2023), the Global Forecast System (GFS) from National Centers for Environmental Prediction (NCEP) (Han and Pan 2011), etc. Krishnan et al. (Krishnan et al. 2023) performed a comprehensive review on solar irradiance forecast, and verified the accuracy of NWP in one-day-ahead solar irradiance over purely data-driven methods. Based on NWP data, accurate PV power forecast can be obtained through irradiance-to-power conversion model like the physical model chain (Yang and Kleissl 2023). Mayer et al. (Mayer et al. 2023) used physical model chain for converting the NWP data of the studied place to the future PV power forecasts, but the physical model chain cannot accurately characterize the complex nonlinear mapping from weather data to PV power. To mitigate the inaccuracy of single physical model chain, Mayer and Yang (Mayer and Yang 2022) used an ensemble of many model chains to convert NWP data to PV power and realize probabilistic forecasts.

Above methods directly map the NWP weather forecasts to PV power forecasts. However, NWP inevitably has intrinsic errors in weather forecasting because of the initial condition, the spatial grid resolution limitation, the inaccurate modelling of atmosphere physics, etc. Thus, above methods can cause errors in PV forecast, and NWP error correction can reduce the errors for PV power forecast to some extent. Researchers have proposed various hybrid methods that combine machine learning and NWP for one-day-ahead PV power forecast. Gneiting et al. (Gneiting et al. 2023) proposed a new method for post-processing NWP data, which can merge statistical techniques with neural networks. Brester et al. (Brester et al. 2023) used a numerical weather prediction data and weather observations to train neural networks, and achieved good PV forecast performance. Conventional neural networks cannot guarantee good performance in extracting the temporal dependency. To characterize the temporal dependency better, Garip et al. (Garip et al. 2023) used long short-term memory (LSTM) network to post-process the NWP of the given place and achieve one-day-ahead PV forecast. Transformer are more advanced technique for characterizing temporal dependency than LSTM and usually have better abilities to extract temporal information. Phan et al. (Phan et al. 2023) proposed a hybrid Transformer-Lower Upper Bound Estimation model and missing data imputation for one-day-ahead probabilistic power forecast using the NWP data of the studied place. Markovics and Mayer (Markovics and Mayer 2022)

performed a comprehensive comparative study on machine learning-based NWP error correction for PV power forecast, compared 24 machine learning models in total, and reported that kernel ridge and multilayer perceptron are the overall most accurate models. Mayer (Mayer 2022) reported that combining machine learning and NWP data can achieve better PV power forecast accuracy than conventional methods through detailed experiments.

Current studies on NWP error correction only use NWP information of the target PV power station, and the spatial NWP information from the neighbouring areas are often neglected. In various fields, such as traffic forecast, nature gas flow forecast etc., researchers have introduced spatial information into forecasts and achieved great success. Jiang et al. (Jiang et al. 2023) proposed a new spatial-temporal meta-graph learning for traffic forecasting and significantly improved the forecast performance. Chen et al. (Chen et al. 2023) proposed a new graph spatial-temporal position recurrent network, characterized spatial-temporal dependency well, and improved the forecast performance. Yang et al. (Yang et al. 2023) used graph neural network to extract the spatial correlation information in natural gas network pipelines. Du et al. (Du et al. 2023) proposed a knowledge-enhanced graph to characterize the temporal dependency of natural gas flow and achieved better forecast performance than conventional methods. Through introducing spatial information and extracting spatial information through deep learning, the forecast performance has been significantly improved than conventional methods.

Although introducing spatial information has significantly improved the forecast accuracy of traffic flow and nature gas flow, spatial information is often neglected in the field of PV power forecasts. Traffic flow and nature gas flow are usually modelled by graph and graph neural network is often used for their forecasting. For NWP correction-based PV power forecasting, NWP data has regular spatial grids, and thus researchers often interpolated to the given station without considering the spatial grid information of NWP. However, current studies focus on using temporal NWP information of the target place. There haven't been relevant studies on whether introducing spatial NWP information in the neighbouring places can bring information gain and improve the accuracy of one-day-ahead PV forecast. Meanwhile, there haven't been any metrics to measure the information gain of introducing spatial NWP information of the neighbouring places in PV forecast field. These are exactly the problems that this paper deals with.

Different from conventional methods, this paper innovatively introduced transfer entropy into PV forecast to reveal the information gain of using NWP information in spatially adjacent areas from the perspective of information theory. Deep spatial multivariable attention CNN-based framework is proposed for one-day-ahead PV power

forecast and achieved significant improvements in forecasting performance.

The main contributions of this paper are summarized as follows:

- (1) **Transfer entropy is innovatively used to measure the spatial NWP information gain in the spatially adjacent places.** To the best of our knowledge, this is the first time that the spatial information gain of introducing NWP information in the neighbouring places in one-day-ahead PV power forecast has been systematically verified from the perspective of information theory.
- (2) **The concept of spatial grid NWP correction for one-day-ahead PV power forecast is innovatively proposed.** Through introducing NWP information from the spatially adjacent places, the concept of spatial grid NWP correction is innovatively proposed for one-day-ahead PV power forecast. The influence of different spatial grid sizes on the one-day-ahead PV forecast performance is also discussed.
- (3) **Deep spatial multivariable attention CNN-based framework is innovatively proposed for one-day-ahead PV power forecast.** Through introducing spatial NWP information and attention mechanism, multivariable spatial-temporal NWP information is adaptively fused and significant improvement in the forecasting performance is achieved.

The rest of this paper is organized as follows. Section "Method" first introduces NWP briefly, then proposes the deep spatial multivariable attention CNN-based spatial grid NWP error method for one-day-ahead PV power forecast. Section "Experiment" performs detailed experiments

to verify the superiorities of the proposed one-day-ahead PV power forecast method. Section "Conclusion and future work" concludes the paper and outlines the future research orientation.

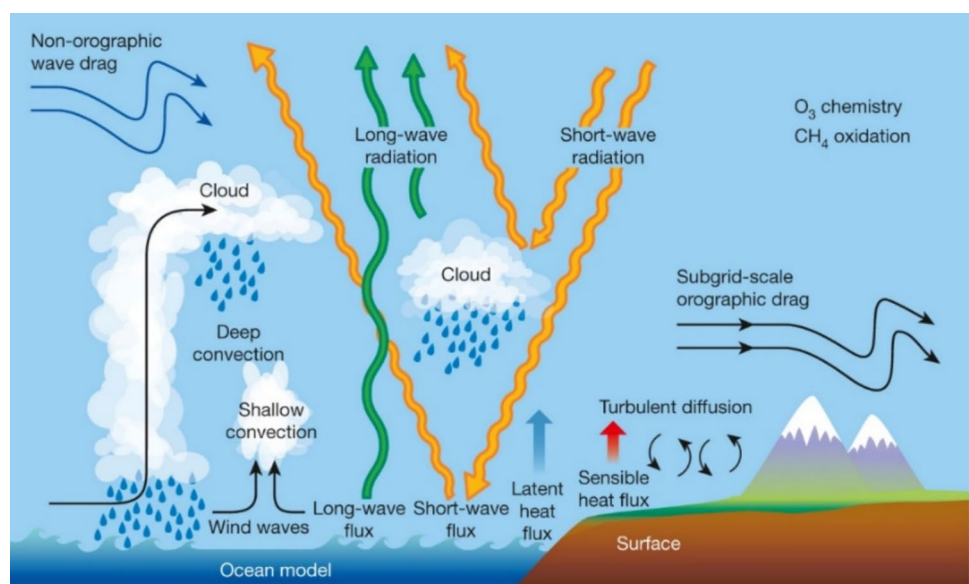
Method

This section will describe the proposed deep spatial multivariable attention convolutional neural network (CNN)-based spatial grid error correction of numerical weather prediction (NWP) framework for one-day-ahead PV power forecast in detail. First, the concept of NWP and the necessities of NWP correction for PV power forecast is briefly introduced in Section "Numerical weather prediction for PV forecast". In Section "Conventional NWP-based one-day-ahead PV forecast method", spatial NWP information is introduced in one-day-ahead PV power forecast, and the concept of spatial grid error correction of NWP is proposed. In Section "Spatial grid error correction of NWP", transfer entropy analysis is used to verify the information gain of introducing spatial NWP information. In Section "Transfer entropy for characterizing information gain", based on the spatial NWP information, the deep spatial multivariable attention CNN-based spatial grid error correction of NWP framework is proposed for one-day-ahead PV power forecast.

Numerical weather prediction for PV forecast

PV power generation are mainly influenced by weather conditions like solar irradiance, air temperature etc. Numerical weather prediction (NWP) theory is the basic theory for weather forecasting. Figure 1 shows the physical principle of

Fig. 1 Physical principle of NWP (Bauer et al. 2015)



NWP (Schultz et al. 2021). Given the weather conditions at the current time, the future weather conditions can be given by integrating a set of partial differential equations (PDEs) as follows (Pu and Kalnay 2018):

$$\begin{aligned}\frac{d\vec{V}}{dt} &= -\alpha\vec{\nabla}p - \vec{\nabla}\Phi + \vec{F} - 2\Omega \times \vec{V} \\ \frac{\partial\rho}{\partial t} &= -\vec{\nabla} \cdot (\rho\vec{V}) \\ p\alpha &= RT \\ Q &= C_p \frac{dT}{dt} - \alpha \frac{dp}{dt} \\ \frac{\partial\rho q}{\partial t} &= -\vec{\nabla} \cdot (\rho\vec{V}q) + \rho(E - C)\end{aligned}\quad (1)$$

where $\vec{V} = (u, v, w)$ is the velocity of air, t is the time, α is the specific volume, ρ is density, p and T are pressure and temperature, Φ is geopotential height, q is the water vapor mixing ratio, Q is heating, E and C represent evaporation and condensation, respectively, R is the gas constant, and $\vec{F}$ is the friction force.

Currently, there are many mature NWP products. Among these products, the High Resolution (HRES) model from European Centre for Medium-Range Weather Forecasts (ECMWF) is generally believed to be one of the most accurate NWP products (Wang et al. 2022a). Therefore, this paper uses ECMWF's HRES model for one-day-ahead PV power forecast.

Literature (Yang et al. 2022) provided an open-source ECMWF's NWP datasets, which include four-year (2017–2020) NWP data in the geographical region bounded by 63°N, −126°W, 21°S, and 36°E (covering most parts of Europe and North America). The dataset provided by literature (Yang et al. 2022) has the spatial resolution of 0.5°*0.5° and the temporal resolution of one hour. The weather variables provided by NWP dataset of literature (Yang et al. 2022) are listed in Table 1, which include solar irradiance, air temperature, wind speed, surface pressure, cloud cover, precipitation etc. This paper used the 18Z NWP data provided by literature (Yang et al. 2022) for one-day-ahead PV power forecasts.

Conventional NWP-based one-day-ahead PV forecast method

The goal of one-day-ahead PV forecast is to obtain the 15-min-interval forecast of the following data. NWP gives the future weather prediction. Through the above open-source ECMWF's NWP dataset provided by literature (Yang et al. 2022), users can obtain the next day's weather forecasts with one-hour temporal resolution and 0.5°*0.5° (about 50 km * 50 km) spatial resolution.

Conventional methods first interpolate NWP to the given station and then uses the physical PV model to convert weather forecasts to solar PV power forecasts. Specifically, conventional methods first use the bilinear interpolation of

Table 1 List of weather variables

Abbr	Long name	Unit
fdir	Total sky direct solar radiation at surface	J/m ²
sp	Surface pressure	Pa
tcwv	Total column water vapor	kg/m ²
u10	10-m U wind component	m/s
v10	10-m V wind component	m/s
t2 m	2-m temperature	K
d2 m	2-m dew point temperature	K
ssrd	Surface solar radiation downwards	J/m ²
strd	Surface thermal radiation downwards	J/m ²
Lcc	Low cloud cover	(0–1)
tco3	Total column ozone	kg/m ²
Tisr	TOA incident solar radiation	J/m ²
Tp	Total precipitation	m
Fal	Forecast albedo	(0–1)

the 0.5°*0.5° spatial grid along both the longitude and latitude directions to obtain the one-day-ahead weather forecasts of the target PV station. Figure 2a shows the bilinear interpolation process. Meanwhile, linear interpolation is also needed to convert the one-hour-interval weather forecasts into 15-min-interval weather forecasts. Then physical PV model is used to convert the weather forecasts into PV power generation forecasts. A commonly-used physical PV model is as follows (Ren et al. 2023):

$$P_S = \frac{H_S}{H_{STC}} \cdot P_{ST} \cdot c_1 [1 + c_2(T_s - T_{STC})] \quad (2)$$

$$T_s = T_a + H_S \cdot \frac{T_{s,TETC} - T_{a,TETC}}{H_{TETC}} \quad (3)$$

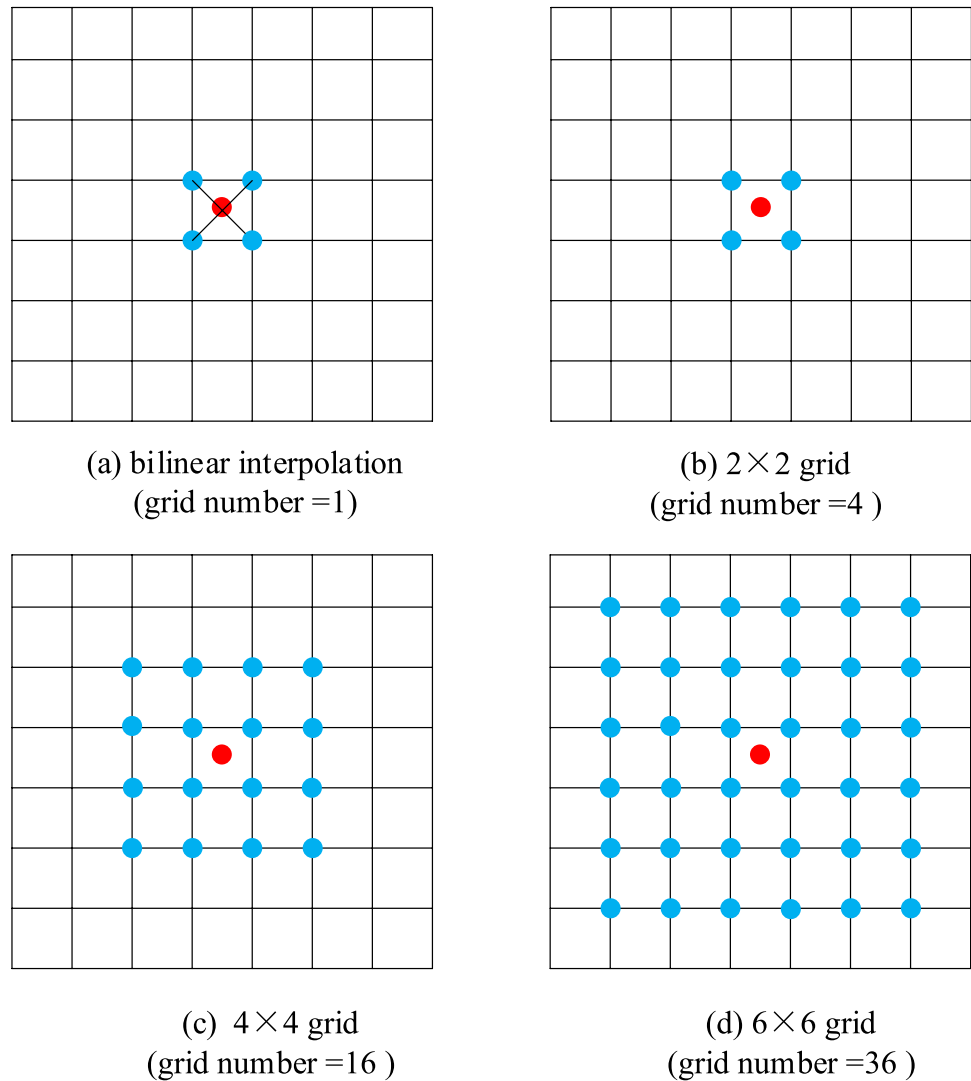
where P_S is the PV power generation. H_S is the solar irradiance and T_a is the temperature. H_{STC} , P_{ST} , c_1 , c_2 , $T_{a,TETC}$, $T_{s,TETC}$ and H_{TETC} are all constants. Specifically, H_{STC} is 1000 W/m², P_{ST} is the rated PV power, T_{STC} is 25 °C, $T_{s,TETC}$ is 47°C, $T_{a,TETC}$ is 20°C, H_{TETC} is 800 W/m², c_1 and c_2 are two constant coefficients for a given PV panel.

Through NWP and physical PV model, one-day-ahead PV power generation forecasts can be obtained.

Spatial grid error correction of NWP

Although the above method can give the future PV power forecasts, it suffers from the intrinsic errors of NWP. NWP inevitably derives from the future weather observations due to the accuracy of initial condition, the spatial grid resolution, the modelling accuracy of physical processes in the atmosphere etc. Thus, it is necessary to perform NWP error correction. Specifically, the future weather observations can

Fig. 2 NWP error correction with different number of spatial grids



be viewed as a nonlinear function of NWP forecasts. Meanwhile, as is shown in Eq. (2)-(3), PV power generation is a nonlinear function of weather observations including solar irradiance, air temperature etc. Thus, the future PV power generation can be viewed as a nonlinear function of NWP data given in Table 1, where the nonlinear function can be fitted by deep neural networks.

Conventional NWP correction methods obtain the NWP forecast of the target PV stations, and regard the future PV power generation as a function of NWP forecasts in the target PV stations. However, the NWP forecasts of the target PV stations are usually obtained by bilinear interpolation (Wu et al. 2023), which assumes that the spatial distribution are linear in the neighbouring places and can introduce interpolation errors due to the neglect of the spatial nonlinearity. Therefore, this paper introduces the NWP information of spatially adjacent places and proposes the concept of spatial grid NWP error correction to realize more accurate PV power forecasts.

As is shown in Fig. 2b-d, the spatial NWP information in the neighbouring places is introduced in one-day-ahead PV power forecasts and interpolation is avoided. In Fig. 2b-d, the red dot is the target PV station, and the blue dots are the used NWP forecasts. For Fig. 2b, four blue dots are used to fit the PV power generation of the target PV station (the red dot), and thus the spatial grid number is 4. For Fig. 2c, 16 blue dots are used to fit the PV power generation, and thus the spatial grid number is 16. For Fig. 2d, 36 blue dots are used to fit the PV power generation, and thus the spatial grid number is 36. The spatial grid number represents the size of the selected spatial neighbourhood parameters in spatial grid correction of NWP. All the NWP information in the selected spatial neighbourhood is inputted to the deep neural networks to fit the future PV power generation.

Based on above analysis, this paper proposes spatial grid correction of NWP data. Different from conventional interpolation methods, this paper used the NWP values from a series of spatial grids to fit the PV power, avoided the grid

interpolation errors and thus can improve the forecasting errors. In the following section, this paper will introduce transfer entropy to measure the information gain of introducing spatial NWP information and verify the forecast performance improvements through experiments.

Transfer entropy for characterizing information gain

This section introduces transfer entropy to reveal the spatial information gain from the perspective of information theory. Transfer entropy can measure the information gain of adding new features in forecast. Specifically, transfer entropy is a widely used method to test whether adding the historical values of another correlated time series can help improve the forecast accuracy of the studied time series (Barnett and Bossomaier 2012; Mao and Shang 2017; Abdul Razak and Jensen 2014). Transfer entropy test has been widely used in economics (Akadiri and Ajmi 2020), finance market (Dimpfl and Peter 2013) and neuroscience (Wibral et al. 2014) to analyze the relationships of different variables in forecast, and achieved great success. Literature (Lindner et al. 2019; Barnett et al. 2009) reveals that transfer entropy is equivalent to Granger causality for linear relationship and has better performance for nonlinear relationship. Therefore, we selected transfer entropy for measuring the spatial information gain in PV power forecasting.

Given the time lag L , then the transfer entropy from time series X to time series Y , denoted by $T_{X \rightarrow Y}$, is defined as follows:

$$T_{X \rightarrow Y} = H(Y_t | Y_{t-1:t-L}) - H(Y_t | Y_{t-1:t-L}, X_{t-1:t-L}) \quad (4)$$

where $H(Y_t | Y_{t-1:t-L})$ is the conditional entropy that characterizes the uncertainty of Y_t when L historical samples of time series Y are given. $H(Y_t | Y_{t-1:t-L}, X_{t-1:t-L})$ is the conditional entropy that characterizes the uncertainty of Y_t when L historical samples of time series Y and time series X are both given. If $H(Y_t | Y_{t-1:t-L})$ is much larger than $H(Y_t | Y_{t-1:t-L}, X_{t-1:t-L})$, then transfer entropy is significantly larger than zero. This means that adding the historical values of time series X can effectively reduce the uncertainty of Y_t , and thus help improve the forecast accuracy of time series Y .

Applying transfer entropy to spatial grid NWP grid correction, and the information gain of introducing spatial NWP information can be revealed. Let W and $\tilde{W}$ denote the NWP forecasts of the target PV stations and the NWP forecasts in the spatial neighbouring places, respectively. Meanwhile, let P denote the actual PV power generation. Then, the information gain IG of introducing spatial NWP information can be defined in Eq. (5):

$$IG = H(P|W) - H(P|W, \tilde{W}) \quad (5)$$

If the information gain IG of introducing spatial NWP information is significantly larger than zero, then it means that introducing spatial NWP information can significantly reduce the uncertainty of PV power generation forecast, and improve the one-day-ahead PV power forecast accuracy.

Deep spatial multivariable attention CNN-based PV forecasting framework

After verifying the information gain of introducing spatial NWP information, this paper proposed the new deep spatial multivariable attention convolutional neural network (CNN)-based spatial-temporal super-resolution framework for one-day-ahead PV power forecast.

CNN is one of the most popular deep learning methods and has strong feature extraction ability to deal with image data. Image data is composed of a series of pixels, while NWP data is composed of weather prediction data at each longitude and latitude grid point. This, NWP data is quite similar to image data and CNN is suitable for dealing NWP data. CNN is composed of convolutional layers, pooling layers, fully connected layers etc.

The convolutional layer is the core of CNN, which can extract the image features quite effectively. Mathematically, the output of a convolutional layer is given in Eq. (6).

$$y^j = f(b^j + \sum_i w^{ij} * x^i) \quad (6)$$

where x^i and y^j are the input and output vectors respectively. w^{ij} and b^j are the convolutional kernel and the bias respectively. The symbol $*$ is the convolution operation. $f(\cdot)$ is activation function. CNN usually uses rectified linear unit (ReLU), i.e., $f(x) = \max(0, x)$.

In the proposed spatial grid NWP error correction framework, multivariable NWP information of the studied place and the spatially adjacent places need to be extracted. The importance of different weather variables and the NWP information from different places in one-day-ahead PV power forecast are different. Adaptive information fusion of multivariable spatial NWP information is needed for accurate PV power forecast. Attention mechanism is one of the most popular methods for adaptive information fusion, which is like people's brains and gives larger weights to more important features in the deep neural networks. Thus, attention mechanism is introduced to adaptively fuse the multivariable spatial NWP information in PV power forecast. When applying the attention mechanism, for each forecast step, a "context vector" c_i and an attention weight a_{ij} are computed as follows:

$$a_{ij} = \frac{f(h_i, s_j)}{\sum_{j=1}^k f(h_i, s_j)} \quad (7)$$

$$c_i = \sum_{j=1}^k a_{ij} s_j \quad (8)$$

where k represents the input sequence length, h_i is the hidden state of decoder at time i , s_j is the hidden state of encoder at time j , $f(\cdot)$ is the presupposed similarity function and a_{ij} is the attention weight for adjusting the proportion of NWP information from different weather variables and different places. The attention weight a_{ij} can be adaptively adjusted through the backpropagation in the network training process, and thus the adaptive information fusion of multivariable spatial NWP information can be realized.

Applying attention mechanism and CNN to the spatial grid NWP correction-based PV power forecast, this paper proposed the deep spatial multivariable attention CNN-based spatial-temporal forecasting framework. The network input includes 15 weather variables, namely 14 NWP weather forecast variables in Table 1 and the clear-sky solar irradiance (Stein et al. 2012). Each weather variable has the spatial grid of 10×10 and the grid resolution of 0.5° . Thus, the network input dimension is $n \times 15 \times 10 \times 10$, where n is batch size, 15 means that there are 15 variables in total, 10×10 represents the grid size of longitude and latitude. The network output is the PV power generation of the target station at current time. The output dimension is $n \times 1$, where n is batch size. The network architecture is shown in Fig. 3. Detailed parameters of input attention module and spatial feature attention module are given in Table 2.

In Fig. 3, 15 weather variables with the spatial grid of 10×10 are first processed by the input attention to adaptively

assign feature importance weights, and then two convolutional layers (Conv1 and Conv2 in Fig. 3) are used for feature extraction. Spatial feature attention is adopted to adaptively assign different weights and achieve spatial information fusion. Finally, one convolutional layer (Conv3 in Fig. 3) is used to further extract the information, followed by the flatten layer and fully-connected layers to give the PV power generation forecast of the target PV station. For Conv1, Conv2 and Conv3, Rectified Linear Unit (ReLU) i.e., $f(x) = \max(0, x)$, is used as the activation function (see Table 2).

This paper designed the input attention module and spatial feature attention module to adaptively assign different weights according to feature importance. For the input attention module, it is a convolutional layer whose input dimension and output dimension are the same. For detailed parameters, see Table 2. Through input attention module, the attention weight matrix $W_{attention}$ is obtained, which has the same dimension as the input matrix X_{input} . Element-wise product, denoted by $\odot$, is adopted to compute the attention output $X_{attention}$ as follows:

$$W_{attention} = \text{Conv}(X_{input}) \quad (9)$$

$$X_{attention} = X_{input} \odot W_{attention} \quad (10)$$

where $\text{Conv}(g)$ is the convolution operation, $X_{attention} \in \mathbb{R}^{n \times 15 \times 10 \times 10}$, $X_{input} \in \mathbb{R}^{n \times 15 \times 10 \times 10}$, $W_{attention} \in \mathbb{R}^{n \times 15 \times 10 \times 10}$ and n is batch size. Through $\text{Conv}(g)$ operation, a learnable attention weight matrix $W_{attention}$ is obtained, where different weather variables can be assigned different weights

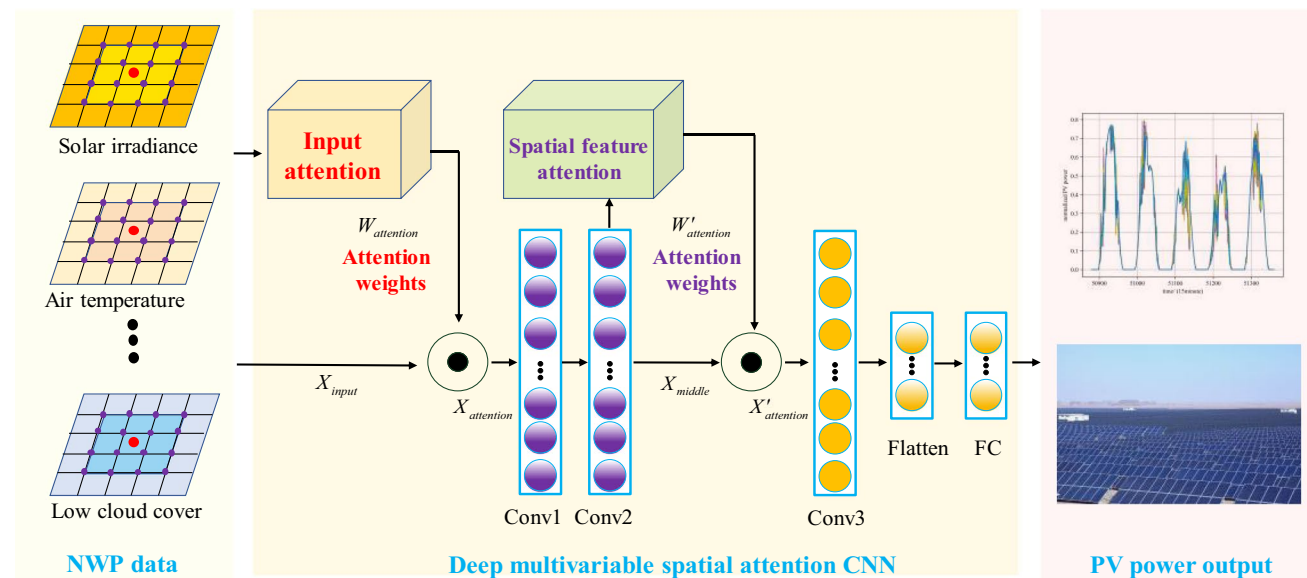


Fig. 3 Deep spatial multivariable attention CNN-based spatial grid NWP error correction framework

Table 2 Detailed inputs and outputs for deep spatial multivariable attention CNN

Layer name	Input shape	Output shape	Parameter details
Input attention	(n, 15, 10, 10)	(n, 15, 10, 10)	Convolutional layer, Kernel number = 15, Kernel size = 1, Padding = 0, Activation function = None
Conv1	(n, 15, 10, 10)	(n, 30, 10, 10)	Convolutional layer, Kernel number = 30, Kernel size = 1, Padding = 0, Activation function = ReLU
Conv2	(n, 30, 10, 10)	(n, 30, 10, 10)	Convolutional layer, Kernel number = 30, Kernel size = 1, Padding = 0, Activation function = ReLU
Spatial feature attention	(n, 30, 10, 10)	(n, 30, 10, 10)	Convolutional layer, Kernel number = 30, Kernel size = 1, Padding = 0, Activation function = None
Conv3	(n, 30, 10, 10)	(n, 30, 10, 10)	Convolutional layer, Kernel number = 30, Kernel size = 1, Padding = 0, Activation function = ReLU
Flatten	(n, 30, 10, 10)	(n, 3000)	None
FC	(n, 3000)	(n, 1)	Fully connected layer, Activation function = None

according to their importance in forecasting. Each element in attention weight matrix $W'_{attention}$ are multiplied with corresponding elements in input matrix X_{input} . Through input attention module, important features are given large weights to help achieve better forecasting performance.

The spatial feature attention module is also a convolutional layer with the same input and output dimension. The output feature matrix X_{middle} of Conv2 layer (see Fig. 3) is processed by convolution operation $Conv'(\cdot)$ to obtain the attention weight matrix $W'_{attention}$. Thus, the attention weight matrix $W'_{attention}$ is a learnable variable related to the matrix X_{middle} , and can be learned through backpropagation during network training. Element-wise product, denoted by $\odot$, is adopted to compute the attention output $X'_{attention}$ as follows:

$$W'_{attention} = Conv'(X_{middle}) \quad (11)$$

$$X'_{attention} = X_{middle} \odot W'_{attention} \quad (12)$$

where $Conv'(g)$ is the convolution operation, $X'_{attention} \in \mathbb{R}^{n \times 30 \times 10 \times 10}$, $X_{middle} \in \mathbb{R}^{n \times 30 \times 10 \times 10}$, $W'_{attention} \in \mathbb{R}^{n \times 30 \times 10 \times 10}$ and n is batch size.

Through training the proposed deep spatial multivariable attention CNN, a nonlinear mapping from the future weather forecasts given by NWP to the future PV power

forecasts can be learned. NWP can give the day-ahead weather forecasts. Through the learning mapping, accurate day-ahead PV power forecasts can be obtained.

Experiment

This section will perform detailed one-day-ahead PV power forecast experiments using real-world data. Section "Data description" will introduce the data used in the experiments. Section "Transfer entropy-based spatial information gain analysis for NWP error correction" will use transfer entropy to verify the information gain of introducing spatial NWP information from the perspective of information theory. Section "Experiment of the proposed spatial grid NWP error correction" will perform detailed one-day-ahead PV power forecast experiments using the proposed deep spatial multivariable attention CNN-based spatial grid NWP error correction framework. Section "Comparison with conventional point NWP correction methods" compares the proposed method with a series of conventional methods to verify its superiorities.

Data description

Three-year actual grid-connected PV power data in Belgium are used in experiments, which are downloaded at <https://www.elia.be/en/grid-data/power-generation/solar-pv-power-generation-data>. The used PV power data are from January 1, 2017 to December 31, 2019 with a temporal resolution of 15 min, namely 96 datapoints per day. The spatial distribution of the studied 11 provinces is shown in Fig. 4. This paper selects Brussels and Flemish Brabant as the studied places. The rated capacity for Brussels and Flemish Brabant is 48.59MW and 253.36MW respectively. Figure 5a and b show the PV power data of the two studied places. The NWP data from ECMWF are used for day-ahead PV power. Figure 6a and b show the one-day-ahead solar



Fig. 4 Spatial distribution of PV stations in Belgium

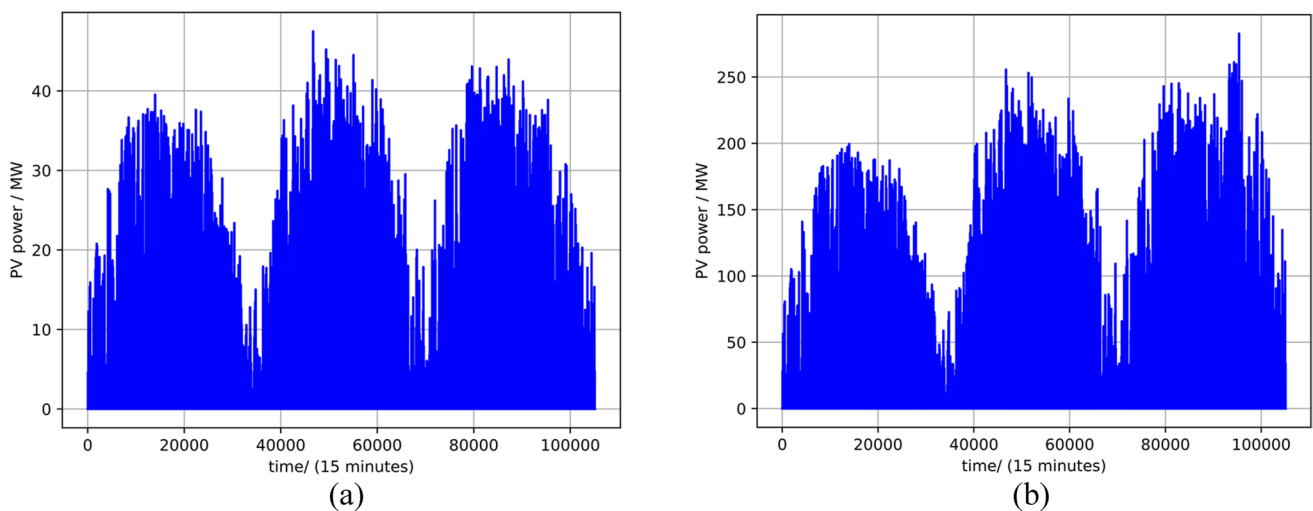


Fig. 5 a PV power in Brussels. b PV power in Flemish Brabant

irradiance forecast given by HRES model's 18Z forecast of ECMWF. Note that we used linear interpolation to convert the ECMWF's one-hour-resolution forecasts into 15-min-resolution solar irradiance forecasts shown in Fig. 6a and b. The reason for selecting two datasets for evaluation is that we want to verify whether the proposed method can obtain the smallest forecasting errors than conventional methods in both Brussels dataset and Flemish Brabant dataset. Through verifying the superiorities of the proposed method in two different datasets, the value of the proposed method can be fully elaborated.

The first two-year (2017–2018) data are used as the training set. The data from the year 2019 are used as the test set. Normalized Mean Absolute Error (nMAE) and normalized Root Mean Squared Error (nRMSE), defined as follows, are used to evaluate the forecast performance.

$$nMAE = \frac{1}{ny_{rated}} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (13)$$

$$nRMSE = \frac{1}{y_{rated}} \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (14)$$

where n is the number of samples, y_i is the actual PV power, $\hat{y}_i$ is the predicted PV power and y_{rated} is the rated PV power or the rated capacity of the given PV power station.

Transfer entropy-based spatial information gain analysis for NWP error correction

Transfer entropy is used to analyze the information gain of introducing spatial grid information. As is shown in Fig. 2, different number of spatial grids are introduced to

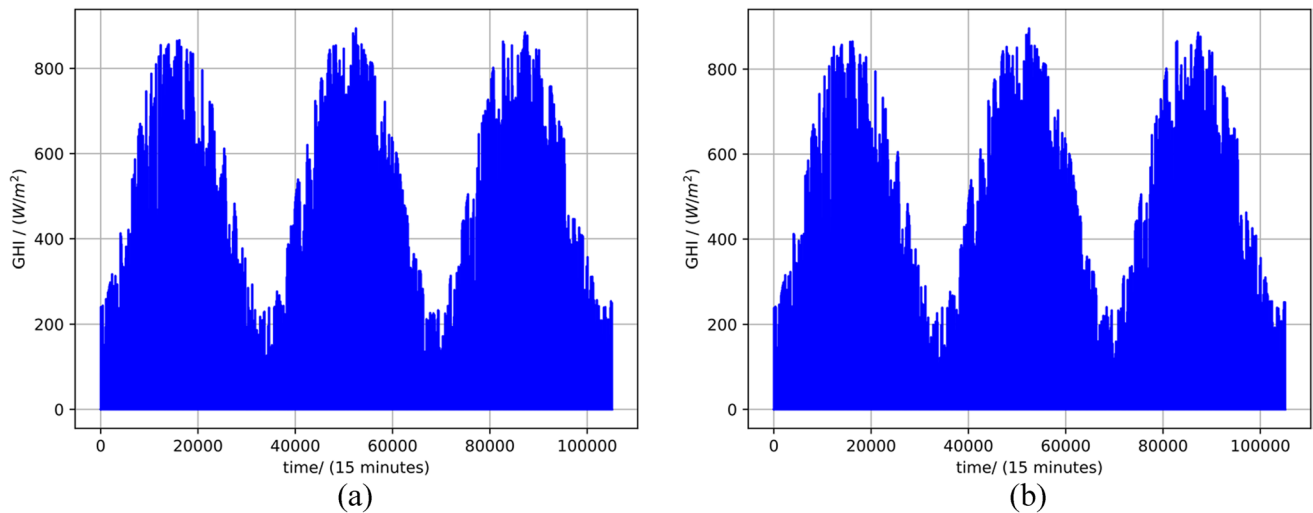


Fig. 6 **a** Forecasted GHI given by ECMWF in Brussels. **b** Forecasted GHI given by ECMWF in Flemish Brabant

Table 3 Transfer entropy analysis result

	Brussels			Flemish Brabant		
	Point entropy	Spatial grid entropy	Information gain percentage	Point entropy	Spatial grid entropy	Information gain percentage
Spatial grid 2 * 2	3.0309	2.8497	5.98%	4.3962	4.0139	8.70%
Spatial grid 4 * 4		1.6768	44.68%		2.2663	48.45%
Spatial grid 6 * 6		0.8793	70.99%		1.1603	73.61%
Spatial grid 8 * 8		0.4706	84.47%		0.6324	85.61%
Spatial grid 10 * 10		0.2434	91.97%		0.3168	92.79%

improve the PV forecast accuracy. This paper computed the point entropy, namely $H(P|W)$ in Eq. (5), and the spatial grid entropy, namely $H(P|W, \tilde{W})$ in Eq. (5). When computing the spatial grid entropy, different spatial grids ranging from 2*2 to 10*10 are used. The information gain percentage is calculated by subtracting the spatial grid entropy from the point entropy and then dividing the result by the point entropy. Experimental results are given in Table 3.

It is observed from Table 3 that spatial grid entropy is significantly smaller than point entropy. This means that introducing spatial grid NWP information can significantly reduce the forecast uncertainty and thus improve the PV forecast accuracy.

Finally, the computational time of transfer entropy under the situation of different spatial grids is discussed. When the grid size increases, the computational complexity also increases. For Brussels dataset, the computational time for 10*10 grid is about 113.95 times larger than the computational time for 1*1 grid. For Flemish Brabant dataset, the computation time increased about 121.65 times. From 1*1 grid to 10*10 grid, the grid size increases 100 times, and the

Table 4 Running time of computing transfer entropy (unit: second)

Grid size	Brussels	Flemish Brabant
Spatial grid 1	0.0109	0.0107
Spatial grid 2 * 2	0.0378	0.0397
Spatial grid 4 * 4	0.1725	0.2060
Spatial grid 6 * 6	0.4499	0.4992
Spatial grid 8 * 8	0.8018	0.8508
Spatial grid 10 * 10	1.2421	1.3017

computational time increases a little larger than 100 times. This means that transfer entropy is scalable when the spatial grid size increases (Table 4).

Experiment of the proposed spatial grid NWP error correction

Attention CNN-based spatial grid error correction of NWP

This section performed experiments of the proposed deep spatial multivariable attention CNN-based spatial-temporal

super-resolution framework. The network is implemented by PyTorch and trained with a single NVIDIA GeForce RTX 3090 graphical card. The parameters during network training are listed in Table 5.

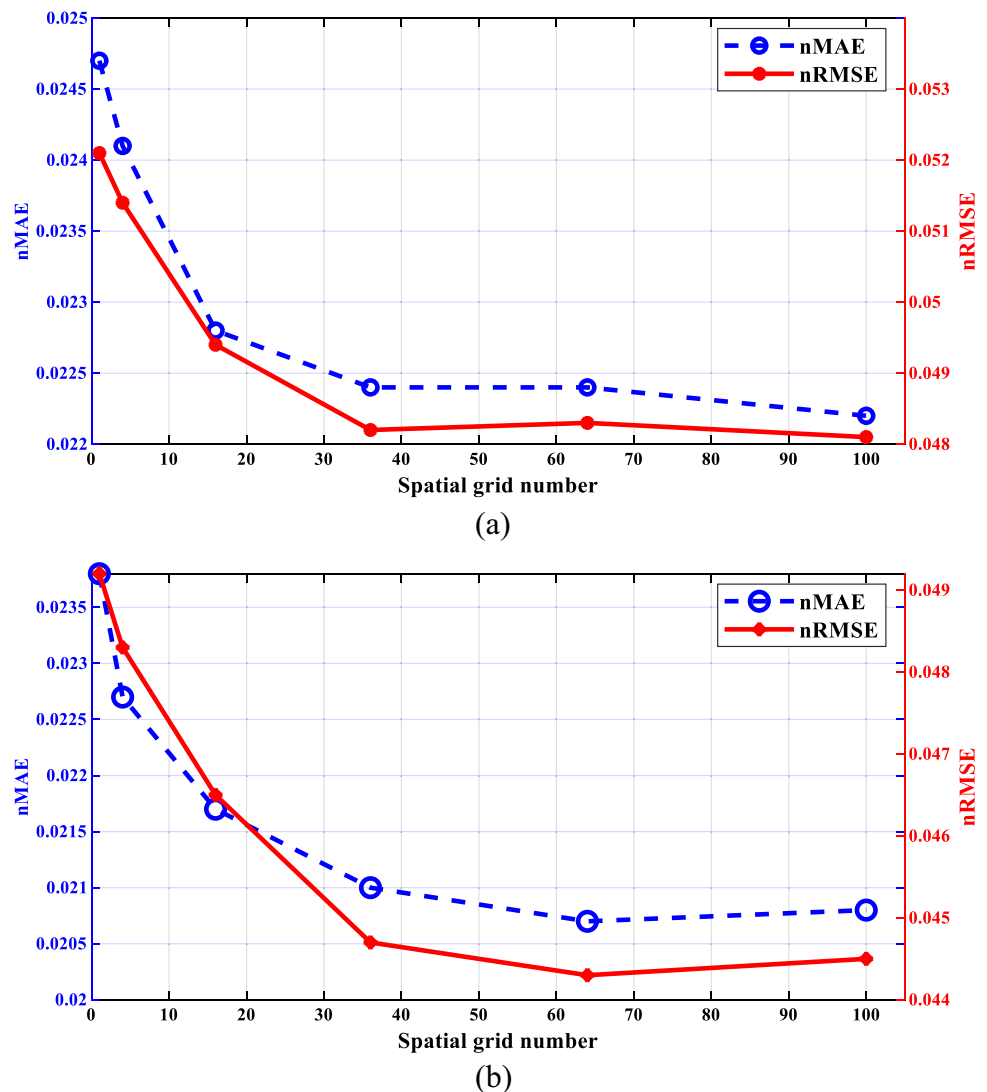
Table 5 The parameters during network training

Parameter	Value
learning rate	0.00005
batch size	64
optimizer	Adam
loss function	mean squared error (MSE)
Train size	Two-year data
Test size	One-year data
Training epoch	100

Different number of spatial grids ranging from 2×2 to 10×10 are used in the proposed multivariable spatial attention CNN to study the influence of spatial grid number on the PV power forecast accuracy. Experiments are performed on the two studied PV stations. The change of forecast errors nMAE and nRMSE with the increase of spatial grid number is given in Fig. 7a and b.

From Fig. 7a and b, it can be observed that forecast errors initially decrease significantly, then stabilize as the number of spatial grids increases. This suggests that increasing the spatial grid resolution can substantially reduce one-day-ahead PV power forecasting errors in the early stages. When the number of spatial grids reaches 10×10 (i.e., 100), the forecast errors for both studied PV stations become nearly constant. Therefore, this paper selects 10×10 as the optimal spatial grid number.

Fig. 7 **a** The change of forecast errors with spatial grid number in Brussels. **b** The change of forecast errors with spatial grid number in Flemish Brabant



Correlation verification between CNN forecast errors and transfer entropy

In Section "Transfer entropy-based spatial information gain analysis for NWP error correction", this paper revealed the spatial information gain through transfer entropy analysis and computed the information gain percentage. In Section "Attention CNN-based spatial grid error correction of NWP", this paper performed detailed experiments to analyze the influence of spatial grid numbers on the forecast errors. Therefore, it is necessary to reveal the relationship between the information gain percentage obtained by transfer entropy analysis and the forecast error reduction. It is also necessary to reveal the relationship between entropy and forecasting errors.

Firstly, this paper analyzed the relationship between entropy and forecasting errors. This paper computed the correlation coefficient between entropy, nMAE and nRMSE when the grid size is [1, 4, 16, 36, 64, 100], and gave the heatmaps to show the correlation in Fig. 8.

It is observed from Fig. 8 that entropy has strong positive correlation with forecasting error measured by nMAE and nRMSE. Entropy is effective in measuring the information uncertainty. Large entropy represents large information uncertainty in forecasting, and thus the forecasting error is usually large in this situation. Thus, entropy has strong positive correlation with forecasting errors.

Next, this paper analyzed the relationship between transfer entropy and forecasting error reduction. This paper used the nMAE reduction and the nRMSE reduction to measure the forecast error reduction, where the nMAE reduction or nRMSE reduction was the difference between the nMAE or nRMSE of using only the NWP information from the target PV stations and the nMAE or nRMSE of using spatial grid information as well. To verify that transfer entropy can reveal the spatial information gain, this section performed Pearson correlation analysis between the information gain percentage and the forecast error reduction. The Pearson correlation coefficient is ranging from -1 to 1 , while the value of -1 represents perfect negative correlation, the value of 0 represents no correlation and the value of 1 represents perfect positive correlation. The results of Pearson correlation analysis in the studied two PV stations are shown in Fig. 9a and b.

From Fig. 9a and b, it is observed that Pearson correlation coefficient is more than 0.96 for both two studied PV stations, and thus forecast error reduction has significant positive correlation with the information gain percentage. This verifies the effectiveness of transfer entropy in revealing the information gain of introducing spatial grid NWP information.

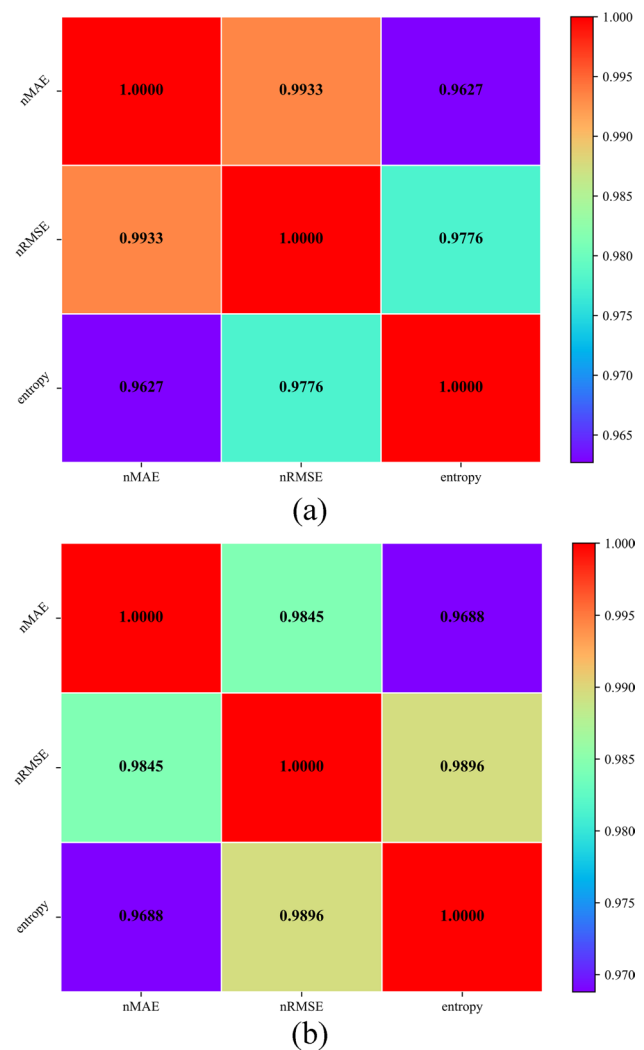


Fig. 8 **a** Correlation between PV power forecasting error and information entropy in Brussels. **b** Correlation between PV power forecasting error and information entropy in Flemish Brabant

Comparison with conventional point NWP correction methods

Comparison with physical PV models

The fundamental methods of one-day-ahead PV power forecasts with NWP is the physical PV model that converts the future weather prediction given by NWP to PV power generation through physical mechanism equations. A simple model that converts solar irradiance and other weather conditions to PV power generation is given in Eq. (2) and Eq. (3), which regards PV power generation as a function of solar irradiance and temperature. Additionally, more accurate method for physical modelling of PV power usually involves a comprehensive modeling chain, including decomposition, transposition, reflection, cell

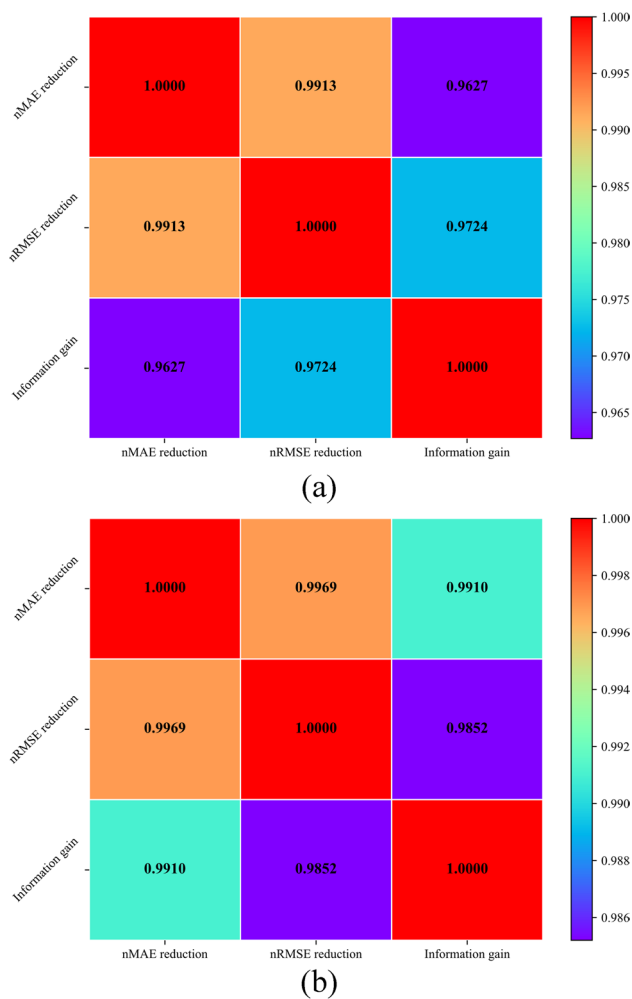


Fig. 9 **a** Correlation between error reduction percentage and information gain percentage in Brussels. **b** Correlation between error reduction percentage and information gain percentage in Flemish Brabant

temperature, PV module, and inverter models. Yang et al. (Yang et al. 2024) gave a comprehensive tutorial review of physical model chain. Wang et al. (Wang et al. 2022b) used physical model chain and NWP data for day-ahead solar PV power forecasting. This paper follows the guideline of literature (Yang et al. 2024; Wang et al. 2022b), and used a series of models including solar position modeling, separation modeling, transposition modeling, cell temperature modeling and PV system modeling to convert NWP to solar PV power. The comparison results among the proposed method, simple PV model and physical model chain are given in Table 6.

From Table 6, the proposed method significantly outperforms simple PV model and physical model chain. Physical model chain has better forecasting performance than simple PV model due to the accurate modeling of all physical procedures of irradiance-to-power conversation. The proposed method effectively extracts the spatial information from the

Table 6 Comparison with physical PV models in Brussels and Flemish Brabant

	Brussels		Flemish Brabant	
	nMAE	nRMSE	nMAE	nRMSE
Simple PV model	0.0348	0.0637	0.0335	0.0610
Model Chain	0.0292	0.0570	0.0276	0.0557
Proposed Method	0.0222	0.0481	0.0208	0.0445

neighboring grids, and thus obtains the best forecasting performance (Fig. 10).

Comparison with conventional machine learning methods

This section compared the proposed multivariable spatial attention CNN-based spatial-temporal method with conventional machine learning methods that only use temporal information of the given station to verify its superiorities. Eight conventional machine learning methods, including Linear Regression (LR), Light Gradient Boosting Machine (LGBM) (Yu and Huo 2022), Random Forest (RF) (Hu and Szymczak 2023), Extreme Learning Machine (ELM) (Huang et al. 2004), Gradient Boosting Decision Tree (GBDT) (Sun et al. 2020), K-Nearest Neighbours (KNN) (Huang et al. 2023), Multilayer Perceptron (MLP) (Altay and Varol 2023), Support Vector Regression (SVR) (Smola and Schölkopf 2004), were compared with the proposed method.

All the above benchmark methods except MLP are implemented by scikit-learn library of Python programming language. MLP is implemented by Keras library of Python programming language. For the hyperparameter tuning of these benchmark methods, this paper used grid search to select the optimal hyperparameters.

Linear Regression (LR) is a linear method without any hyperparameter. LR only has the weight and basis parameter, the two parameters can be determined through the least square method, and thus LR does not need hyperparameter tuning.

For Light Gradient Boosting Machine (LGBM), this paper selected the learning rate from {0.001, 0.01, 0.1} and the number of estimators from {1, 10, 100, 1,000, 10,000}. The optimal hyperparameters for LGBM are a learning rate of 0.01 and 1,000 estimators.

For Random Forest (RF), this paper selected the maximum depth from {1, 10, 100, 1,000} and the number of estimators from {1, 10, 100, 1,000}. The optimal hyperparameter for RF are a maximum depth of 100 and 100 estimators.

For ELM, this paper selected the hidden neuron number from {1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100}, the optimal hidden neuron number is 50 and sigmoid function was used as the activation function.

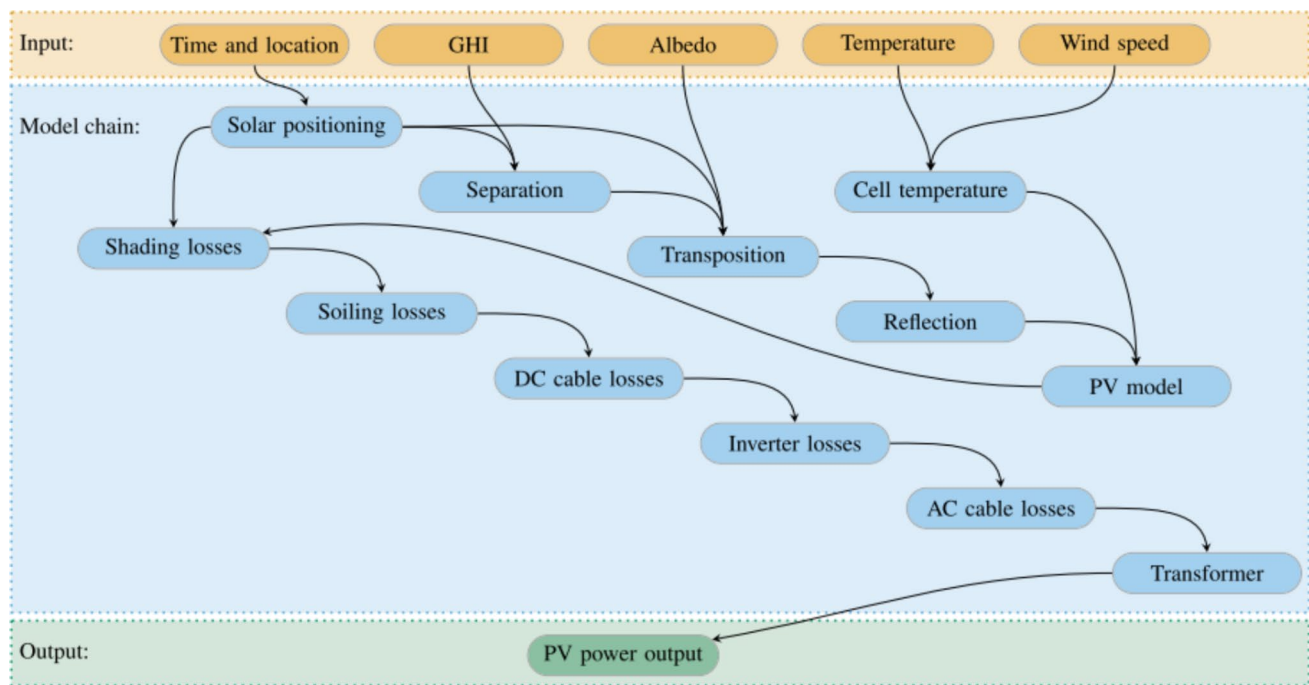


Fig. 10 Principle of physical model chain (Yang et al. 2024)

For GBDT, this paper selected the number of estimators from {1, 10, 100, 1,000, 10, 000} and the optimal hyperparameter is 100.

For KNN, this paper selected the number of neighbors from {1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100} and the optimal hyperparameter is 50.

For MLP, this paper selected the learning rate from {0.001, 0.01, 0.1} and the hidden neuron number from {1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100}. The optimal hidden neuron number for MLP is 50 and the optimal learning rate for MLP is 0.01.

For SVR, this paper selected the parameter C from {1, 10, 100, 500, 1, 000, 10, 000} and the parameter gamma from {0.0001, 0.01, 0.1, 0.5}. The optimal parameter C for SVR is 500 and the optimal parameter gamma for SVR is 0.1.

This paper compared the forecasting performance in Brussels and Flemish Brabant. The experimental comparison results are shown in Table 7.

It is observed from Table 7 that the proposed method obtained the smallest nMAE and nRMSE in the two studied PV station. Thus, the proposed spatial grid NWP correction method significantly improves the one-day-ahead PV power forecast accuracy when compared with conventional methods that only consider the temporal NWP information of the given station. This verifies the effectiveness of introducing spatial NWP information in the neighboring grids in day-ahead PV power forecasting.

To demonstrate the superiorities of the proposed method better, this paper computed the performance improvement

Table 7 Overall forecast errors in Brussels and Flemish Brabant

	Brussels		Flemish Brabant	
	nMAE	nRMSE	nMAE	nRMSE
LR	0.0279	0.0571	0.0267	0.0542
LGBM	0.0249	0.0525	0.0238	0.0500
RF	0.0259	0.0552	0.0245	0.0521
ELM	0.0254	0.0521	0.0241	0.0490
GBDT	0.0260	0.0535	0.0247	0.0505
KNN	0.0279	0.0566	0.0265	0.0535
MLP	0.0259	0.0525	0.0246	0.0497
SVR	0.0277	0.0540	0.0259	0.0499
Proposed Method	0.0222	0.0481	0.0208	0.0445

percentage of the proposed method and the compared eight conventional machine learning methods than physical PV models. The definition of performance improvement percentage is as follows:

$$\delta_{nMAE} = \frac{nMAE_{physical} - nMAE}{nMAE_{physical}} \times 100\% \quad (15)$$

$$\delta_{nRMSE} = \frac{nRMSE_{physical} - nRMSE_{st}}{nRMSE_{physical}} \times 100\% \quad (16)$$

where $nMAE$ and $nRMSE$ are nMAE and nRMSE of the proposed method, and $nMAE_{physical}$ as well as $nRMSE_{physical}$ are

nMAE and nRMSE of the physical PV model that converts weather forecast to PV power generation through physical mechanism equations.

Figure 11 shows the performance improvement percentage of the proposed method and eight conventional machine learning methods. The proposed method achieves the largest performance improvement percentage in both Brussels and Flemish Brabant. When compared with physical model chain, the proposed method reduces nMAE by 23.97% and nRMSE by 15.54% in Brussels. In Flemish Brabant, the proposed method achieves even better results with nMAE being reduced by 24.79% and nRMSE being reduced by 20.19%. These improvements verify the superiorities of the proposed

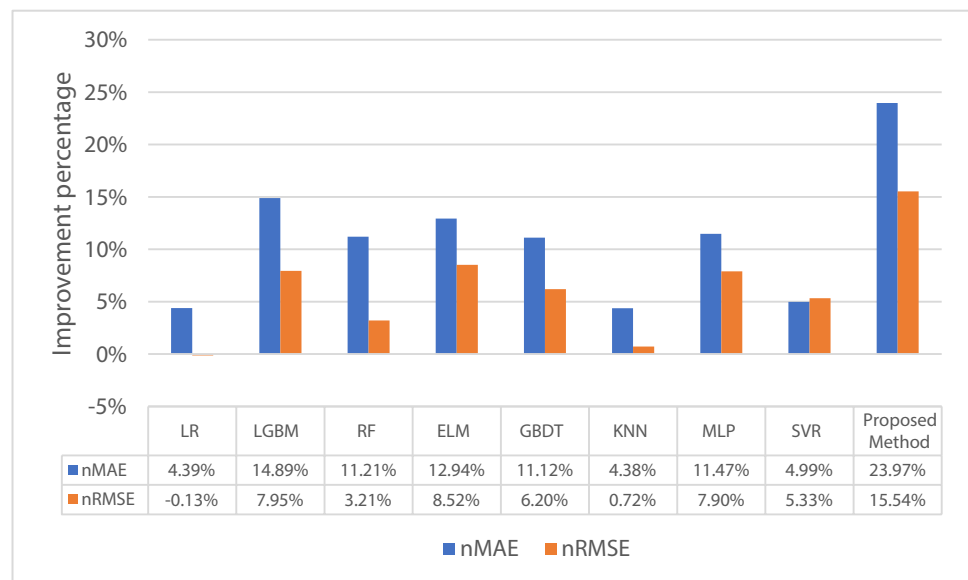
spatial grid NWP correction method for one-day-ahead PV power forecast.

Forecast error comparison in different seasons

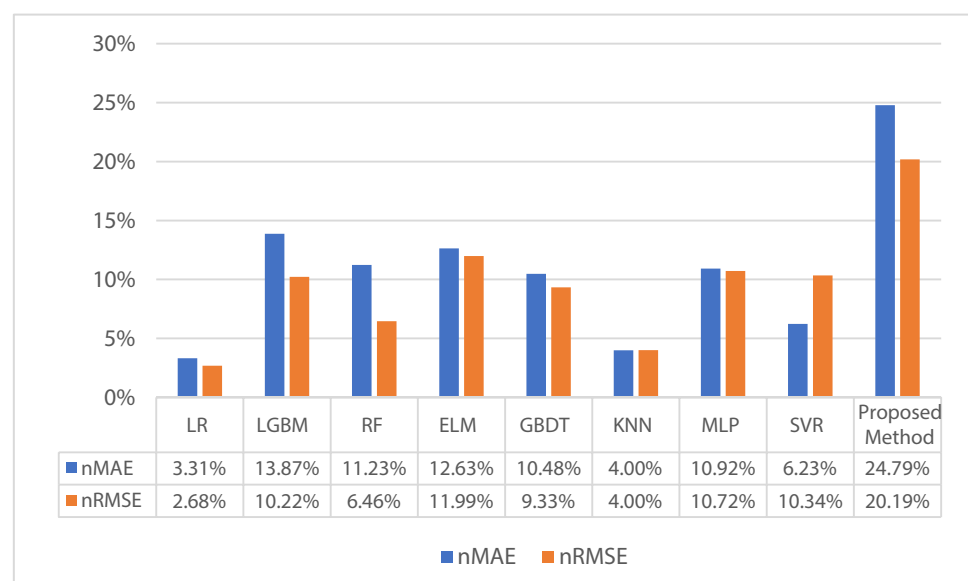
After verifying the significant improvements of the proposed method over conventional methods, this paper computed the nMAE and nRMSE in spring, summer, autumn and winter of the test set, namely the year 2019. The experimental results in Brussels and Flemish Brabant are shown in Tables 8 and 9.

From Tables 8 and 9, it is observed that the proposed method can significantly improve the forecast accuracy in

Fig. 11 **a** Improvement percentage than Physical Model Chain in Brussels. **b** Improvement percentage than Physical Model Chain in Flemish Brabant



(a)



(b)

Table 8 Forecast errors of different seasons in Brussels

	Spring		Summer		Autumn		Winter	
	nMAE	nRMSE	nMAE	nRMSE	nMAE	nRMSE	nMAE	nRMSE
Simple PV model	0.0441	0.0747	0.0478	0.0763	0.0274	0.0541	0.0193	0.0425
Model Chain	0.0376	0.0684	0.0385	0.0656	0.0241	0.0507	0.0160	0.0372
LR	0.0356	0.0672	0.0372	0.0660	0.0227	0.0498	0.0157	0.0401
LGBM	0.0326	0.0643	0.0337	0.0616	0.0204	0.0456	0.0122	0.0308
RF	0.0342	0.0678	0.0348	0.0645	0.0213	0.0477	0.0130	0.0327
ELM	0.0322	0.0621	0.0344	0.0619	0.0204	0.0448	0.0142	0.0335
GBDT	0.0337	0.0642	0.0356	0.0638	0.0209	0.0462	0.0131	0.0322
KNN	0.0358	0.0678	0.0369	0.0663	0.0233	0.0491	0.0153	0.0364
MLP	0.0328	0.0622	0.0348	0.0627	0.0208	0.0449	0.0145	0.0338
SVR	0.0347	0.0647	0.0365	0.0624	0.0229	0.0466	0.0164	0.0366
Proposed Method	0.0289	0.0581	0.0294	0.0562	0.0188	0.0430	0.0112	0.0289

Table 9 Forecast errors of different seasons in Flemish Brabant

	Spring		Summer		Autumn		Winter	
	nMAE	nRMSE	nMAE	nRMSE	nMAE	nRMSE	nMAE	nRMSE
Simple PV model	0.0421	0.0713	0.0462	0.0734	0.0264	0.0515	0.0188	0.0413
Model Chain	0.0330	0.0613	0.0350	0.0648	0.0271	0.0568	0.0148	0.0340
LR	0.0312	0.0634	0.0326	0.0635	0.0193	0.0465	0.0114	0.0384
LGBM	0.0312	0.0643	0.0326	0.0616	0.0193	0.0456	0.0114	0.0308
RF	0.0323	0.0638	0.0329	0.0618	0.0202	0.0445	0.0122	0.0304
ELM	0.0306	0.0581	0.0325	0.0585	0.0194	0.0418	0.0135	0.0318
GBDT	0.0321	0.0606	0.0338	0.0604	0.0200	0.0434	0.0125	0.0305
KNN	0.0337	0.0634	0.0348	0.0628	0.0224	0.0470	0.0147	0.0346
MLP	0.0314	0.0591	0.0332	0.0597	0.0196	0.0420	0.0138	0.0320
SVR	0.0325	0.0595	0.0353	0.0595	0.0210	0.0424	0.0144	0.0323
Proposed Method	0.0278	0.0551	0.0275	0.0517	0.0176	0.0393	0.0097	0.0248

all seasons of the year for both studied PV stations. It is observed that all the methods have the smallest forecasting errors in winter. Literature (Chen et al. 2020) also reveals similar results that the forecasting error is the smallest in winter and explains it from the physical mechanism of weather systems. In winter, large-scale, long-lived weather systems affect the cloud evolutions the most. For example, when a synoptic frontal system passes the site, it usually is featured with overcast sky over a few days. Then the high-pressure system comes with entire clear-sky or little cloud coverage over a few days. Therefore, winter weather is mostly governed by a large-scale synoptic system that could last several days. This significantly enhances the predictability of winter weather, and thus can account for the reason why the forecasting error in winter is the smallest among all seasons.

Furthermore, the performance improvement percentage than the conventional physical PV model was also computed for the four seasons of the year in the two studied PV stations. The performance improvement percentage for

Brussels and Flemish Brabant is shown in Figs. 12 and 13 respectively.

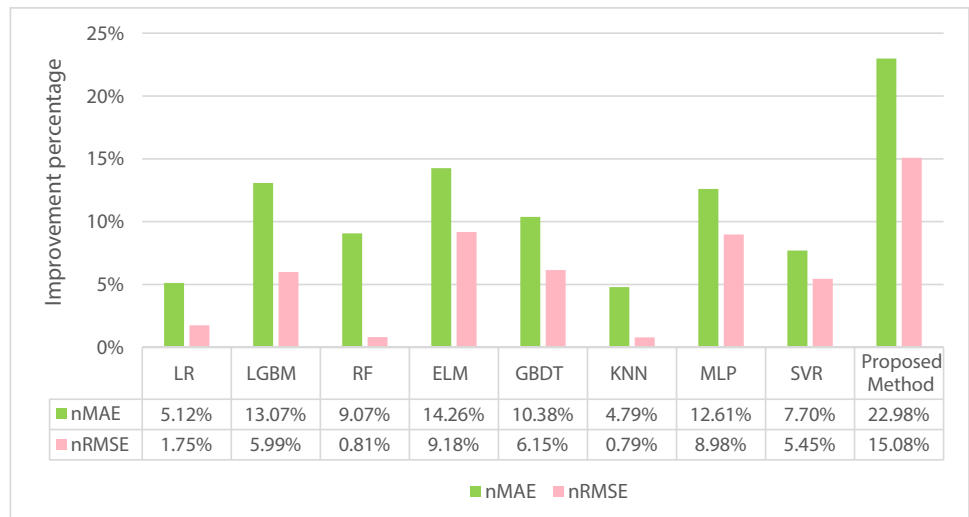
From Figs. 12 and 13, it is observed that the proposed method obtains the largest improvement percentage. Compared with physical model chain, the proposed method enhances nMAE by over 15% during spring, summer, and autumn, and by more than 29% in winter for the two investigated PV stations. Simultaneously, the proposed method improves nRMSE by more than 10% in spring, summer, and autumn, and by more than 22% in winter. This can further verify the superiorities of the proposed method.

Probabilistic forecast experiment

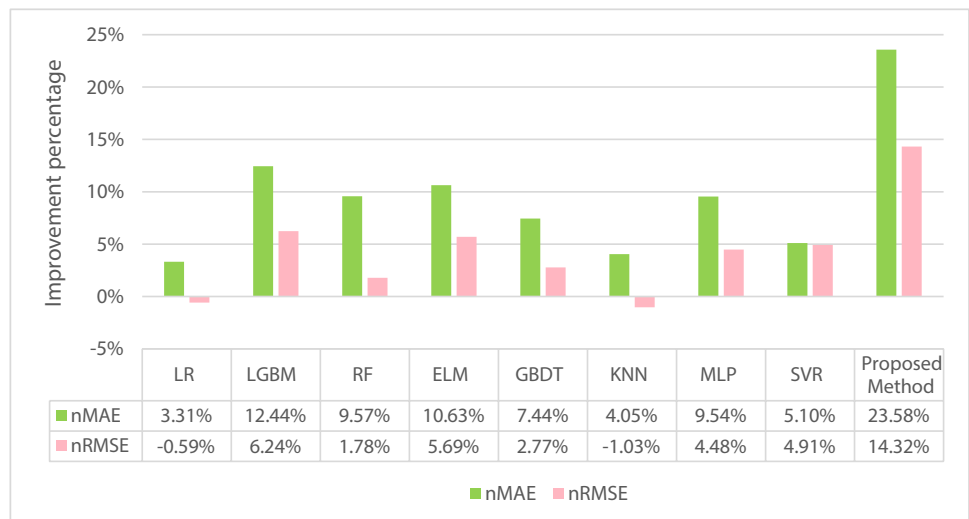
After realizing accurate deterministic one-day-ahead forecasts, it is necessary to give the probabilistic interval forecasts. This paper performed probabilistic interval forecasts through quantile analysis.

For 90% confidence intervals, let the function $f_{95}(\cdot)$ and $f_5(\cdot)$ denote the 95 th quantile and 5 th quantile respectively,

Fig. 12 **a** Improvement percentage than Physical Model Chain of Brussels in spring. **b** Improvement percentage than Physical Model Chain of Brussels in summer. **c** Improvement percentage than Physical Model Chain of Brussels in autumn. **d** Improvement percentage than Physical Model Chain of Brussels in winter



(a)



(b)

then the upper limit and the lower limit of the prediction are defined by Eq. (17)–(19).

$$\hat{u}_{t,upper} = \hat{y}_t + f_{95}(\{e_{train}\}) \quad (17)$$

$$\hat{u}_{t,lower} = \max(\hat{y}_t + f_5(\{e_{train}\}), 0) \quad (18)$$

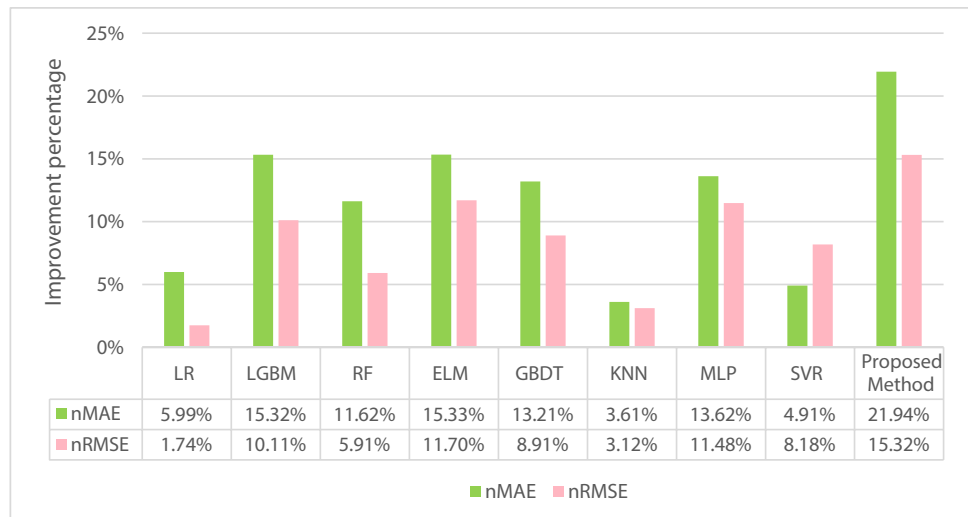
$$e_{train,t} = y_{train,t} - \hat{y}_{train,t} \quad (19)$$

where $\{e_{train}\}$ is the forecast errors of training set. For each training sample, the forecast error $e_{train,t}$ is defined by the actual PV power $y_{train,t}$ in training set minus the predicted PV power $\hat{y}_{train,t}$ in training set. Through computing the 95 th quantile and 5 th quantile of the forecast errors in training set, the upper limit and lower limit of the probabilistic

interval can be obtained by adding 95 th quantile and 5 th quantile to the predicted PV power respectively. Considering the minimal values of PV power generation is zero, $\max(\cdot)$ function is used for the lower limit to avoid negative values. Furthermore, the 80% confidence interval and 70% confidence interval can be obtained through similar methods.

To evaluate the effectiveness of probabilistic PV forecast, prediction interval coverage probability (PICP) and prediction interval normalized average width (PINAW) (Zhao et al. 2021) defined in Eq. (20) and Eq. (21), are used. PICP represents the proportion of samples within the prescribed confidence interval, while PINAW denotes the normalized average width of the confidence intervals. Experimental evaluation results for 90%, 80% and 70% confidence interval are listed in Table 10.

Fig. 12 (continued)



(c)



(d)

$$PICP = \frac{1}{n} \sum_{t=1}^n I_t \quad (20)$$

$$PINAW = \frac{1}{ny_{rated}} \sum_{t=1}^n (\hat{u}_{t,upper} - \hat{u}_{t,lower}) \quad (21)$$

$$I_t = \begin{cases} 1, & \text{if } y_t \in [\hat{u}_{t,lower}, \hat{u}_{t,upper}] \\ 0, & \text{if } y_t \notin [\hat{u}_{t,lower}, \hat{u}_{t,upper}] \end{cases} \quad (22)$$

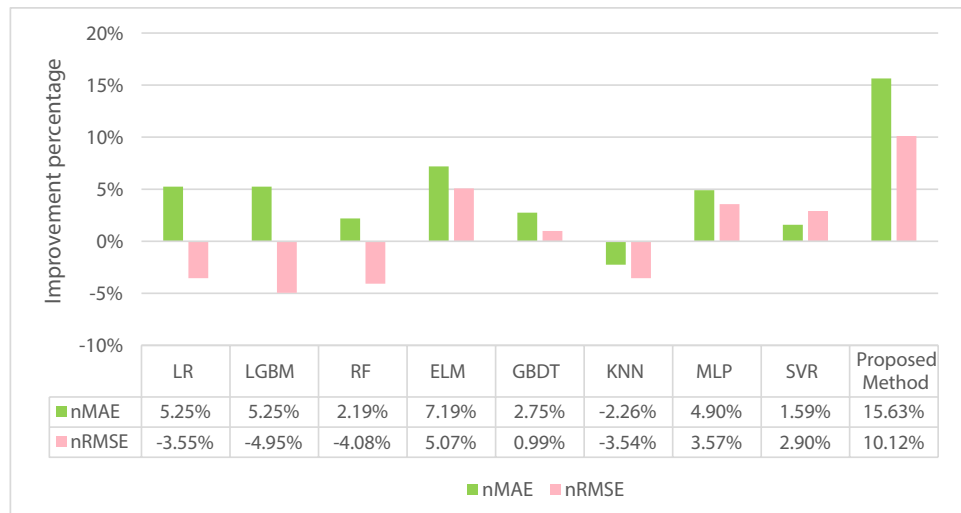
where n is the total number of samples in the test set, y_t is the actual PV power at time step t , $\hat{u}_{t,upper}$ and $\hat{u}_{t,lower}$ denote the upper and lower limits of the confidence interval at time step t respectively, y_{rated} refers to the rated capacity of the corresponding PV station, and I_t is a binary variable that

indicates whether the actual PV power value at time step t lies within the given confidence interval $[\hat{u}_{t,lower}, \hat{u}_{t,upper}]$.

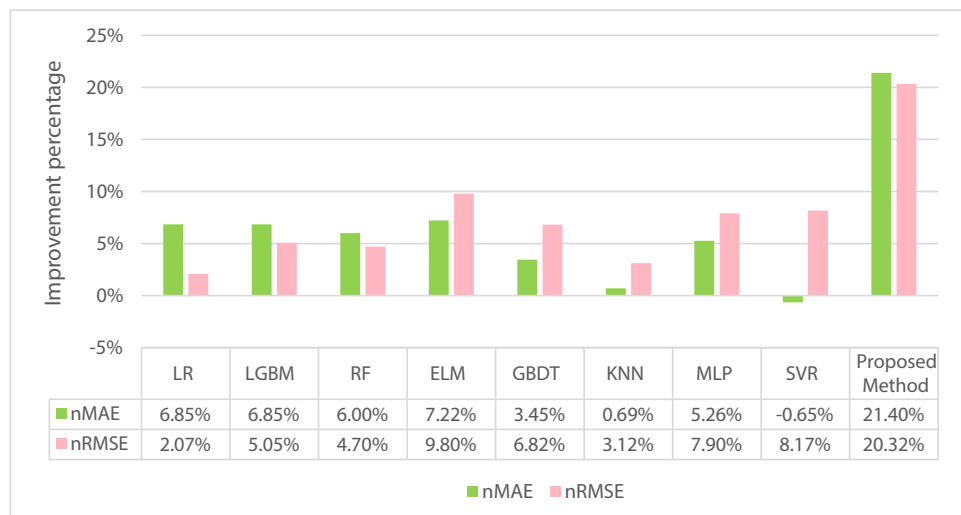
From Table 10, it is observed that the PICP for the 90%, 80%, 70% confidence interval is all close to the nominal values, and the PINAW is relatively narrow, which verifies the effectiveness of the proposed probabilistic PV power forecast method.

To better demonstrate the probabilistic PV power forecast results, the visualization of the probabilistic PV power forecast results is shown in Fig. 14. In Fig. 14, the red line and the yellow line are the predicted PV power and the actual PV power respectively, and the green shadow denotes the 90% confidence interval. It is observed that the red and yellow lines are mostly within the green shadows, which further verify the reliability and effectiveness of the proposed probabilistic PV power forecast method.

Fig. 13 **a** Improvement percentage than Physical Model Chain of Flemish Brabant in spring. **b** Improvement percentage than Physical Model Chain of Flemish Brabant in summer. **c** Improvement percentage than Physical Model Chain of Flemish Brabant in autumn. **d** Improvement percentage than Physical Model Chain of Flemish Brabant in winter



(a)



(b)

After performing probabilistic interval forecasting on the proposed method, this paper also performed probabilistic interval forecasting on benchmark methods including simple PV model, physical model chain and eight conventional machine learning methods. PICP and PINAW metrics are also used to evaluate the probabilistic interval forecasting performance of the benchmark methods. Comparison results are listed in Table 11.

From Table 11, it is observed that the PICP for the 90%, 80%, 70% confidence interval is all close to the nominal values, which shows that the proposed probabilistic forecasting method is effective for all the compared methods. Meanwhile, the proposed method has the smallest PINAW value among all the compared methods. This means the proposed method has significantly smaller interval width than all the other compared methods. The proposed method can obtain the smallest forecasting error for deterministic point

forecasting and the smallest probabilistic interval width for probabilistic interval forecasting, which verifies the significant superiorities of the proposed method.

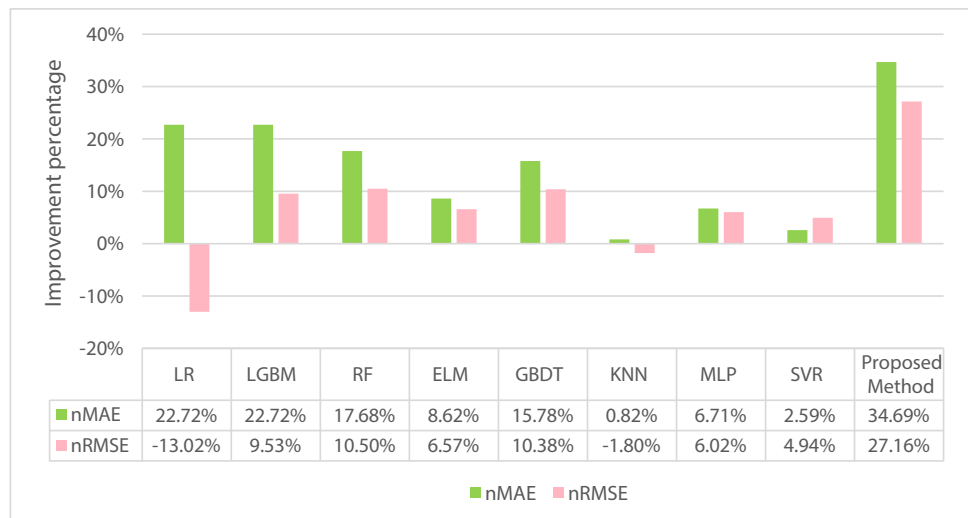
Forecasting performance evaluation on distribution-based metrics

In the above section, this paper evaluated the deterministic point forecasting using nMAE and nRMSE, and evaluated the probabilistic interval forecasting using PICP and PINAW. Although these metrics have proven the superiorities of the proposed method, these evaluation metrics do not capture the distributional characteristics of predictions. To strengthen the evaluation, this section introduces additional distribution-based metrics for evaluation. This section introduces Kullback–Leibler divergence (KLD) (Cui et al. 2024), Kolmogorov–Smirnov (KS) test (Ohunakin

Fig. 13 (continued)



(c)



(d)

Table 10 Probabilistic forecast result

		Brussels		Flemish Brabant	
		PICP	PINAW	PICP	PINAW
90%	train	90.00%	0.0694	90.00%	0.0654
	test	89.77%	0.0694	89.57%	0.0654
80%	train	80.00%	0.0393	80.00%	0.0366
	test	79.93%	0.0393	79.18%	0.0366
70%	train	70.00%	0.0221	70.00%	0.0211
	test	69.63%	0.0221	69.25%	0.0211

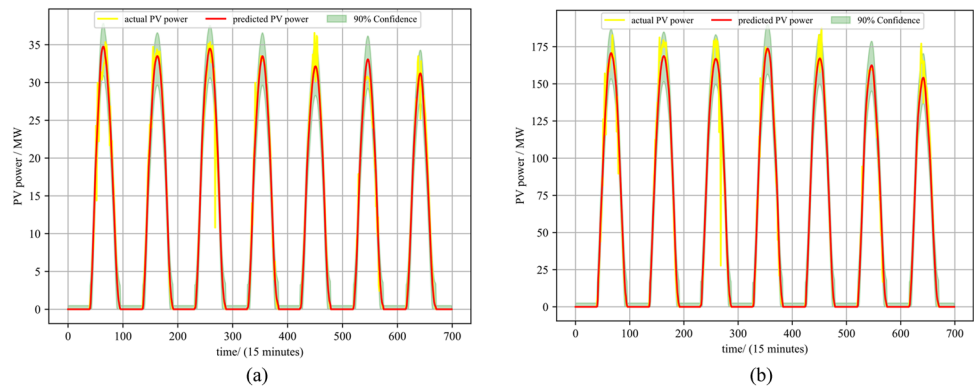
et al. 2024), and cumulative distribution function (CDF) analysis for evaluating the deterministic point forecasting. Meanwhile, this section introduces continuous ranked

probability score (CRPS) (Koochali et al. 2025) and sharpness (Koochali et al. 2025) for evaluating the probabilistic interval forecasting.

Kullback–Leibler divergence (KLD) measures the differences between two probability distributions. For distributions P and Q of a continuous random variable, it is defined as follows (Cui et al. 2024):

$$D_{KL}(P||Q) = \int_{-\infty}^{+\infty} p(x) * \log \frac{p(x)}{q(x)} dx \quad (23)$$

where p and q denote the probability densities of P and Q . If P is the observed distribution and Q is the predicted distribution, $D_{KL}(P||Q)$ measures the information loss when using Q to approximate P . Smaller KLD means better forecasting performance.

Fig. 14 **a** Probabilistic interval of Brussels. **b** Probabilistic interval of Flemish Brabant**Table 11** Probabilistic forecast comparison results

		Brussels			Flemish Brabant		
		90%	80%	70%	90%	80%	70%
Simple PV	PICP	90.12%	79.88%	69.88%	89.89%	79.75%	69.93%
	PINAW	0.1148	0.0736	0.0453	0.1108	0.0720	0.0445
Model chain	PICP	89.37%	79.37%	69.37%	89.37%	79.37%	69.40%
	PINAW	0.0898	0.0574	0.0370	0.0872	0.0504	0.0305
LR	PICP	89.82%	79.66%	69.23%	88.48%	78.26%	68.92%
	PINAW	0.1088	0.0698	0.0430	0.1079	0.0691	0.0429
LGBM	PICP	85.18%	76.23%	67.63%	84.92%	75.96%	67.54%
	PINAW	0.0876	0.0490	0.0260	0.0826	0.0464	0.0251
RF	PICP	88.72%	78.66%	69.93%	88.61%	78.92%	69.69%
	PINAW	0.0805	0.0444	0.0249	0.0767	0.0426	0.0239
ELM	PICP	89.24%	78.71%	68.59%	88.85%	78.67%	68.86%
	PINAW	0.0803	0.0464	0.0265	0.0761	0.0447	0.0260
GBDT	PICP	89.60%	79.14%	69.46%	89.36%	79.24%	69.58%
	PINAW	0.0806	0.0471	0.0274	0.0772	0.0460	0.0268
KNN	PICP	87.44%	77.44%	67.43%	87.29%	77.37%	67.40%
	PINAW	0.0774	0.0459	0.0270	0.0734	0.0441	0.0259
MLP	PICP	90.21%	80.23%	69.63%	89.96%	79.70%	69.69%
	PINAW	0.0838	0.0493	0.0284	0.0794	0.0474	0.0278
SVR	PICP	87.13%	77.54%	67.64%	86.60%	77.02%	67.44%
	PINAW	0.0771	0.0510	0.0328	0.0719	0.0479	0.0309
Proposed	PICP	89.77%	79.93%	69.63%	89.57%	79.18%	69.25%
	PINAW	0.0694	0.0393	0.0221	0.0654	0.0366	0.0211

Cumulative distribution function (CDF) analysis aims to analyze CDF. The Cumulative Distribution Function (CDF) is a fundamental concept in probability theory and statistics that describes the probability that a random variable X takes a value less than or equal to x . Mathematically, it is defined as follows. Through analyzing the CDF of the predicted values and observed values, the similarity between them can be verified.

$$F_X(x) = P(X \leq x) \quad (24)$$

CDF analysis can visualize the distribution and see if two distributions are similar to some extent. To better judge whether

the predicted distribution is similar to the observed distribution, Kolmogorov–Smirnov (KS) test (Ohunakin et al. 2024) is introduced. If p-value is smaller than the given significant level like 0.01, then it is reasonable to believe that there are no significant differences between the predicted distribution and the observed distribution. This means that the predicted values and the observed values have similar distributions.

Continuous Ranked Probability Score (CRPS) is an example of a strictly proper scoring rule for evaluating probabilistic forecasts. For a forecast distribution F and an observed outcome x , the CRPS is defined as (Koochali et al. 2025). Smaller CRPS means better performance.

$$D_{KL}(P\|Q) = \int_{-\infty}^{+\infty} p(x) * \log \frac{p(x)}{q(x)} dx \quad (25)$$

Sharpness measures the concentration of the predictive distribution (Koochali et al. 2025). Following the guideline of literature (Koochali et al. 2025), this paper quantifies sharpness using the ratio of the 95 th percentile (p_{95}) and the 5 th percentile (p_5), denoted by p_{95}/p_5 . For each sample in the test set, p_{95}/p_5 can be computed, and thus there is a series of p_{95}/p_5 values, and the median value of these p_{95}/p_5 values is used to evaluate sharpness and method comparison.

This paper evaluated the forecasting performance of the proposed method and benchmark methods using the above distribution-based metrics in the selected two PV stations. Experimental results are given in Tables 12 and 13. The CDF figures of the two PV stations are shown in Fig. 15.

From Table 12 and Table 13, it is observed that the proposed method has the smallest CRPS, sharpness and KLD among all the compared methods. This means the predicted distribution of the proposed method is closest to the observed distribution among all the compared methods. Meanwhile, the p-value of all the methods is smaller than the given significant level 0.01, which means the predicted distribution is similar to the observed distribution.

Computational time analysis

The consumed time of model inference is important. In real-world applications, users usually train a deep learning model offline, and then perform inference of the trained model to get the model output. The consumed time in actual applications is the model inference time. Thus, this paper evaluated

Table 13 Distribution-based metrics for forecasting comparison in Flemish Brabant

Method	CRPS	Sharpness	KLD	KS test	
				KS Statistic	KS p-value
Simple PV model	0.0264	2.0835	0.0186	0.0885	0.0000
Model chain	0.0225	1.8496	0.0022	0.0130	0.0054
LR	0.0271	2.0302	0.0086	0.0355	0.0000
LGBM	0.0224	1.5619	0.0148	0.0173	0.0001
RF	0.0209	1.7584	0.0097	0.0193	0.0000
ELM	0.0196	1.7905	0.0179	0.0408	0.0000
GBDT	0.0204	1.7540	0.0094	0.0167	0.0001
KNN	0.0216	1.7177	0.0293	0.0189	0.0000
MLP	0.0200	1.8394	0.0147	0.0306	0.0000
SVR	0.0204	1.7849	0.0121	0.0456	0.0000
Proposed Method	0.0176	1.6311	0.0043	0.0151	0.0007

the model inference time. This paper used the trained model to predict the future PV power generation for samples in test set. Experiments were performed through a computer with one RTX 3090 Graphics card running Ubuntu 20.04 operation system. Inference was performed 10 times, and the mean inference time is used as the final consumed time. Experimental results are given in Table 14.

From Table 14, it is observed that the proposed method consumes only 0.0002694 s ~ 0.0002722 s. The PV power data is collected every 15 min. Compared with data collection interval, the consumed time of the proposed method is very small. Thus, the proposed network can be used for real-time online PV power forecasting.

Evaluation on low-resolution NWP data

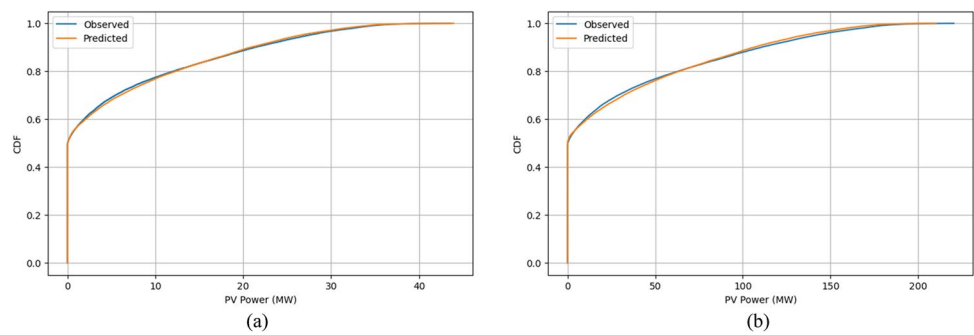
In the above experiments, this paper used the NWP data with the spatial resolution of $0.5^\circ \times 0.5^\circ$. To test the model's performance with low-resolution NWP data, we selected the resolution of $1^\circ \times 1^\circ$ and $1.5^\circ \times 1.5^\circ$. In the experiment, we also used 10×10 spatial grid and the same training parameters as the experiment of $0.5^\circ \times 0.5^\circ$ resolution. The forecasting performance is shown in Table 15.

It is observed that the forecasting error increases in both datasets when the grid resolution becomes lower. The forecasting accuracy reduction percentage is given in Table 16.

It is observed that nMAE increases 5.45% ~ 6.72% and nRMSE decreases 3.89% ~ 4.37% when the grid resolution decreases from $0.5^\circ \times 0.5^\circ$ to $1^\circ \times 1^\circ$. When the grid resolution decreases from $0.5^\circ \times 0.5^\circ$ to $1.5^\circ \times 1.5^\circ$, nMAE increases 9.02% ~ 13.33% and nRMSE increases 7.50% ~ 8.44%. From $0.5^\circ \times 0.5^\circ$ to $1.5^\circ \times 1.5^\circ$, the grid resolution decreases 3 times, but the forecasting errors only increase about 10%.

Table 12 Distribution-based metrics for forecasting comparison in Brussels

Method	CRPS	Sharpness	KLD	KS test	
				KS Statistic	KS p-value
Simple PV model	0.0939	3.2905	0.6830	0.4142	0.0000
Model chain	0.0663	3.7185	0.5800	0.4670	0.0000
LR	0.0807	2.8776	0.0507	0.3053	0.0000
LGBM	0.0558	1.8556	0.0398	0.2292	0.0000
RF	0.0527	1.7704	0.0590	0.3170	0.0000
ELM	0.0594	2.4562	0.0076	0.2630	0.0000
GBDT	0.0554	2.3359	0.0117	0.2593	0.0000
KNN	0.0642	2.1924	0.1568	0.3571	0.0000
MLP	0.0579	3.1716	0.0586	0.3738	0.0000
SVR	0.0592	2.5264	0.0259	0.2640	0.0000
Proposed Method	0.0423	1.6916	0.0032	0.2554	0.0000

Fig. 15 a CDF of Brussels. b CDF of Flemish Brabant**Table 14** Computational time comparison

	Brussels	Flemish Brabant
Mean inference time	0.2722 ms	0.2694 ms

Table 15 PV forecasting performance on different spatial resolutions

Grid resolution	Brussels		Flemish Brabant	
	nMAE	nRMSE	nMAE	nRMSE
0.5°* 0.5°	0.0222	0.0481	0.0208	0.0445
1°* 1°	0.0238	0.0503	0.0220	0.0463
1.5° * 1.5°	0.0244	0.0520	0.0240	0.0486

Table 16 Forecasting accuracy reduction percentage compared with 0.5°* 0.5° resolution NWP data

Grid resolution	Brussels		Flemish Brabant	
	nMAE	nRMSE	nMAE	nRMSE
1°* 1°	6.72%	4.37%	5.45%	3.89%
1.5° * 1.5°	9.02%	7.50%	13.33%	8.44%

This shows that the proposed method is relatively robust to the grid resolution and can still obtain relatively good forecasting performance even with low-resolution NWP data.

New case study of day-ahead PV power forecasting in USA

To verify the generalizability of the proposed method, this paper tested its forecasting performance beyond Europe to validate its broader applicability. This paper selected a PV station in San Diego, USA, whose location is 32.45°N, 117.10°W. The used PV power data is provided in literature (Silwal et al. 2021) and can be downloaded at <https://github.com/sushilsilwal3/UCSD-Microgrid-Database>. The rated power of Alternating Current (AC) for the selected PV station is 35kW. Three-year data from year 2017 to year 2019 was used in the experiment, where the data of

Table 17 Forecast error comparison in San Diego, USA

	nMAE	nRMSE
Simple PV model	0.0935	0.1604
Model chain	0.0875	0.1524
LR	0.0909	0.1468
LGBM	0.0588	0.1141
RF	0.0564	0.1142
ELM	0.0702	0.1282
GBDT	0.0653	0.1187
KNN	0.0676	0.1245
MLP	0.0676	0.1245
SVR	0.0711	0.1242
Proposed Method	0.0479	0.0951

year 2017 and 2018 was used as the training set and the data of the year 2019 was used as the test set. Experimental comparison results for deterministic point forecasting and probabilistic interval forecasting are given in Table 17 and Table 18, respectively.

From Tables 17 and 18, it is observed that the proposed method obtains the smallest nMAE and nRMSE for deterministic point forecasting, and the smallest PINAW for probabilistic interval forecasting among all the compared methods.

Furthermore, this paper also used the distributed-based evaluation metrics, namely CRPS, sharpness, KLD, KS test, to evaluate the forecasting performance in San Diego, USA. Experimental results are shown in Table 19.

From Table 19, it is observed that the proposed method obtains the smallest CRPS and sharpness, and the second smallest KLD among all the compared methods. This means that the predicted distribution of the proposed method is very close to the observed distribution, and thus the superiorities of the proposed method is verified.

To sum up, the proposed method obtains the best forecasting performance in both Europe and North America. This verifies the superiorities of the proposed method in different geographical regions and weather patterns.

Table 18 Probabilistic forecast comparison in San Diego, USA

		90%	80%	70%
Simple PV	PICP	90.00%	80.01%	69.85%
	PINAW	0.4509	0.3326	0.2203
Model chain	PICP	88.55%	79.43%	67.79%
	PINAW	0.3353	0.2375	0.1524
LR	PICP	90.32%	79.37%	69.16%
	PINAW	0.4001	0.2749	0.1730
LGBM	PICP	88.75%	79.27%	70.15%
	PINAW	0.2488	0.1525	0.0888
RF	PICP	86.61%	76.65%	67.93%
	PINAW	0.2205	0.1177	0.0620
ELM	PICP	87.77%	78.01%	67.73%
	PINAW	0.2988	0.1811	0.1058
GBDT	PICP	89.31%	79.90%	69.25%
	PINAW	0.2790	0.1723	0.0998
KNN	PICP	89.28%	79.51%	69.49%
	PINAW	0.3076	0.1947	0.1152
MLP	PICP	89.41%	80.61%	69.51%
	PINAW	0.3263	0.2064	0.1183
SVR	PICP	88.27%	78.71%	70.37%
	PINAW	0.3089	0.1851	0.1188
Proposed	PICP	89.77%	79.80%	71.12%
	PINAW	0.1880	0.1088	0.0651

Table 19 Distribution-based metrics for forecasting comparison in San Diego, USA

Method	CRPS	Sharpness	KLD	KS test	
				KS Statistic	KS p-value
Simple PV model	0.0264	3.2905	0.6830	0.4142	0.0000
Model chain	0.0225	3.7185	0.5800	0.4670	0.0000
LR	0.0271	2.8776	0.0507	0.3053	0.0000
LGBM	0.0224	1.8556	0.0398	0.2292	0.0000
RF	0.0209	1.7704	0.0590	0.3170	0.0000
ELM	0.0196	2.4562	0.0016	0.2630	0.0000
GBDT	0.0204	2.3359	0.0117	0.2593	0.0000
KNN	0.0216	2.1924	0.1568	0.3571	0.0000
MLP	0.0200	3.1716	0.0586	0.3738	0.0000
SVR	0.0204	2.5264	0.0259	0.2640	0.0000
Proposed Method	0.0176	1.6916	0.0032	0.2554	0.0000

Conclusion and future work

Accurate one-day-ahead PV power forecast is crucial to the electricity grid dispatching. Conventional one-day-ahead PV power forecast methods only use the NWP information

from the targeted PV station, while the spatial NWP information in the spatially adjacent places is often neglected. To fill this gap, this paper proposed the new concept of the spatial grid NWP correction for one-day-ahead PV power forecast. Transfer entropy were innovatively introduced into one-day-ahead PV power forecasts to reveal the information gain of introducing spatial NWP information in spatially adjacent place from the perspective of information theory. A novel deep spatial multivariable attention CNN-based spatial-temporal super-resolution framework of spatial grid NWP correction was proposed to realize more accurate one-day-ahead PV power forecast.

Through experiments in three-year (2017–2019) actual PV data of two PV stations in Belgium, the superiorities of the proposed method were verified. For one-day-ahead PV power forecast of the two studied PV stations, the proposed method obtains a normalized mean absolute error (nMAE) of 2.08% and 2.22%, along with a normalized root mean squared error (nRMSE) of 4.45% and 4.81%, respectively. Compared with conventional physical method without NWP error correction, the proposed method reduces nMAE by 36.21% ~ 38.02% and nRMSE by 24.40% ~ 27.18%. Comparison with a series of conventional NWP error correction methods further verify the superiorities of the proposed method. Forecast errors were also compared with conventional NWP correction methods in four seasons respectively, which verified the significant superiorities of the proposed methods in all seasons of the year.

Although the proposed method has achieved good PV power forecasting performance, it still has some limitations. In the proposed method, spatial grid NWP information is introduced and the forecasting accuracy is significantly improved. However, this paper does not consider the temporal dependency information in NWP data. Introducing temporal dependency information of NWP and adding temporal attention in the network is promising to further improve the forecasting accuracy in the future.

In the future, the temporal information and temporal attention will be added for more accurate PV power forecasting. Additionally, we will also integrate real-time forecasting, explore hybrid forecasting models, and apply the proposed method to other energy systems like wind power generation, electricity load demand etc.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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